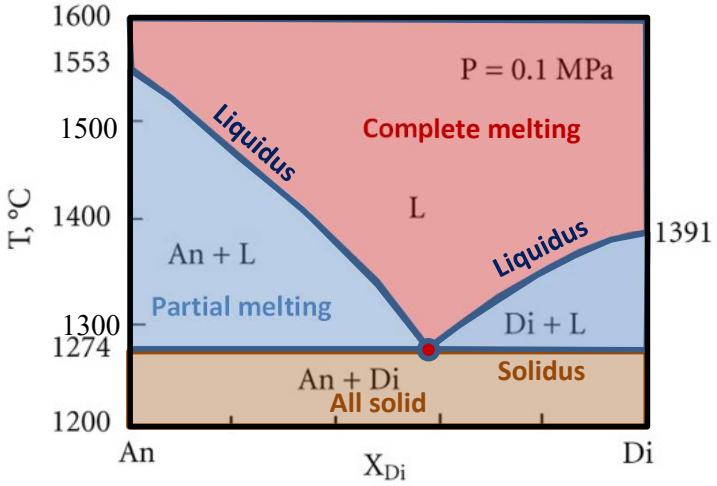
Chemistry of the Earth: Lecture 7
Melting and the Rare Earth Elements (REE)

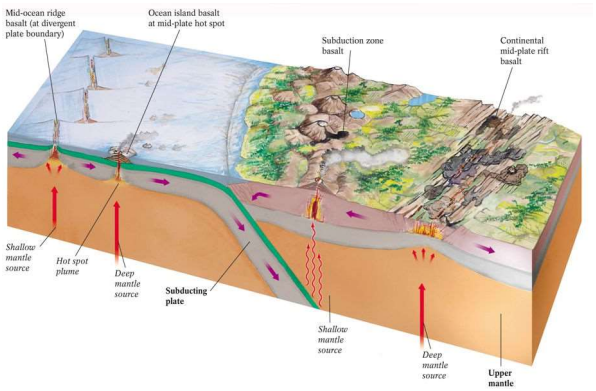


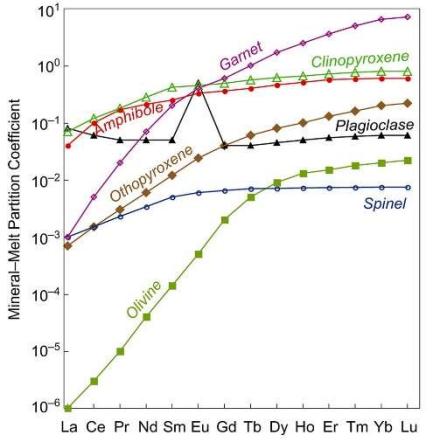
Part 1: How rocks melt

OBJECTIVES OF THIS LECTURE AND PRACTICAL

- Investigate how and why rocks melt
- Use phase diagrams to understand how rocks melt and crystallize over a range of temperatures
- Investigate the relationship between rock chemistry and melt chemistry, and the concept of element partitioning
- Understand how and why the rare earth elements act as powerful tracers of geochemical processes







READING AND RESOURCES

Grotzinger & Jordan (Basic: Minimum reading)	**p.97-98 (melting), p.133 (global volcanism)**
Gill (Recommended)	**Chapter 2, p. 26-30**
Misra	**Chapter 12.3**
White	Chapter 7 - particularly 7.2.2.4, 7.3, 7.4.4
McSween & Richardson	Chapter 12, p. 237 - 255

Internet resources

Rare Earth Elements: Minerals, Mines, Magnets (and More)
http://www.elementsmagazine.org/archivearticles/e8_5/e8_5.pdf - pages 15-22

Additional resource

Diopside-Anorthite.pdf in Blackboard folder

HOW AND WHY DO ROCKS MELT?

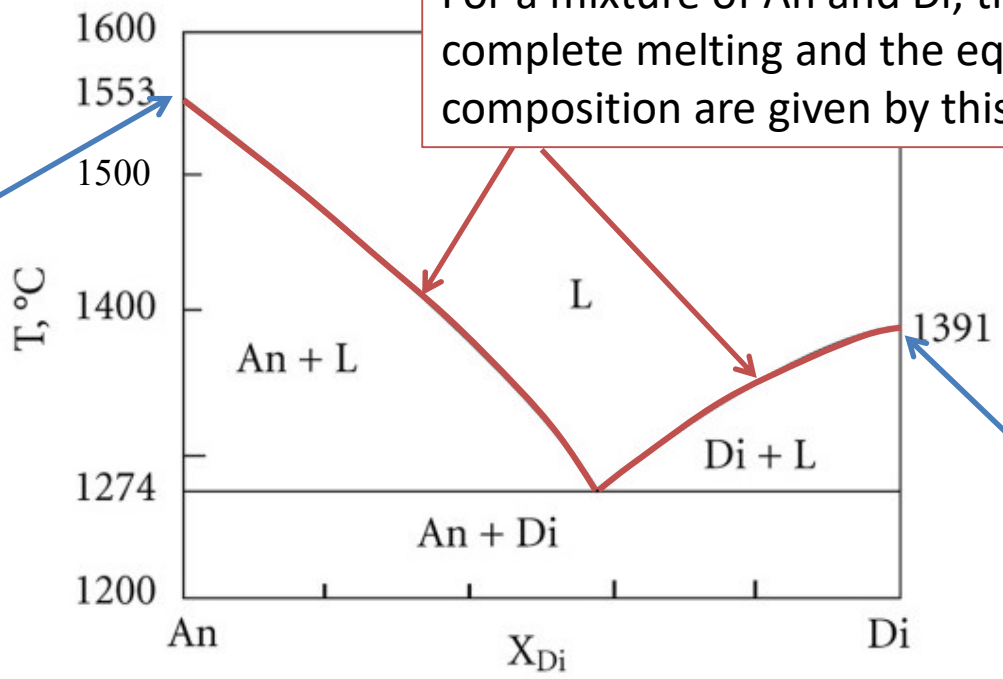
Why? – Rocks begin to melt when the Gibbs free energy of formation of a liquid within the system is less than the Gibbs free energy of the minerals that comprise the rock:

$$\Delta G_{melt} < \Sigma \Delta G_{minerals}$$

How? – Rocks do not have a single "melting point". They melt over a range of temperatures. The equilibrium composition of the melt at a given temperature and pressure is determined by thermodynamics.

For a mixture of An and Di, the temperature of complete melting and the equilibrium melt composition are given by this line, called the liquidus

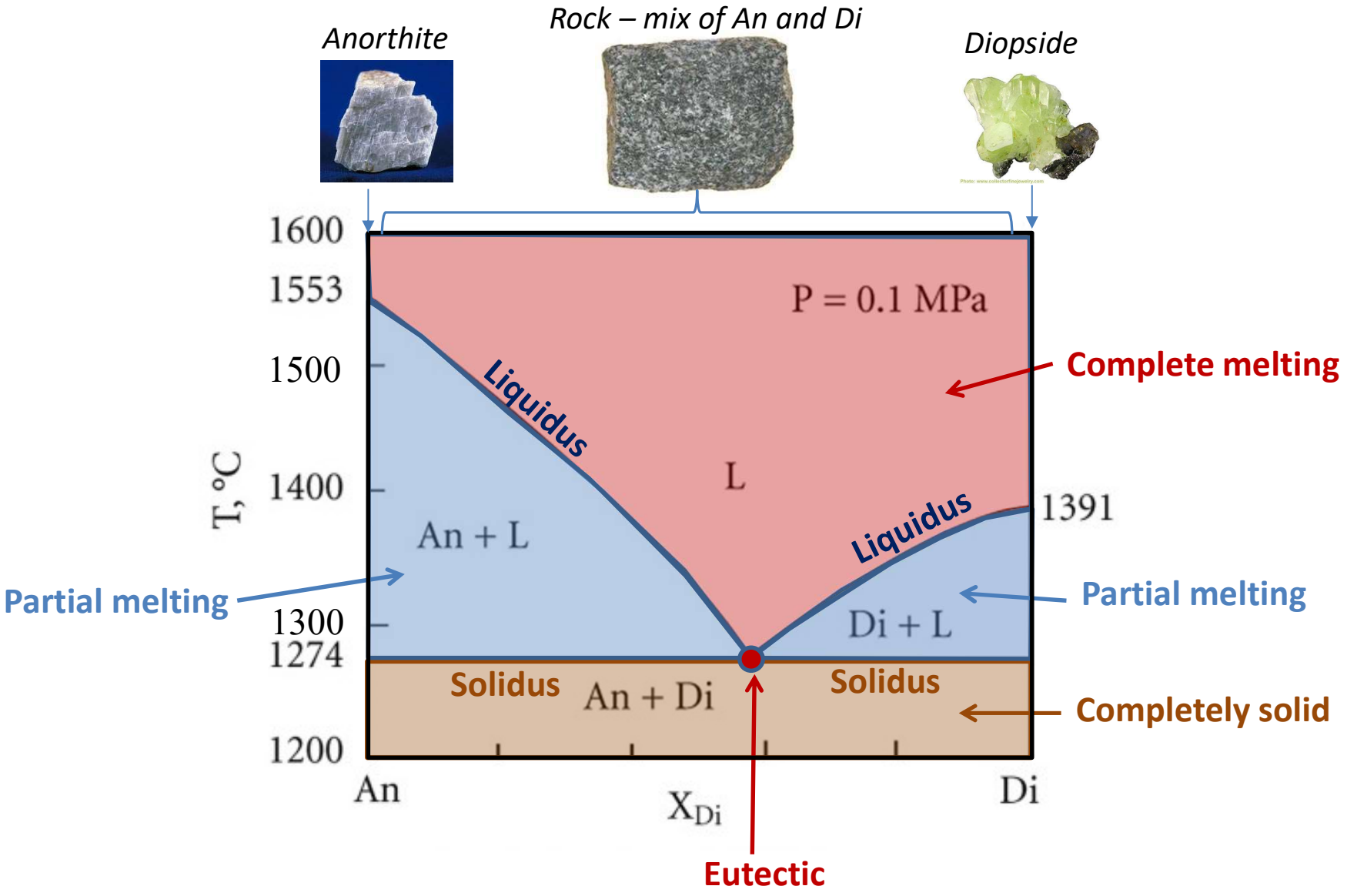
Anorthite – melting point, 1553 °C.



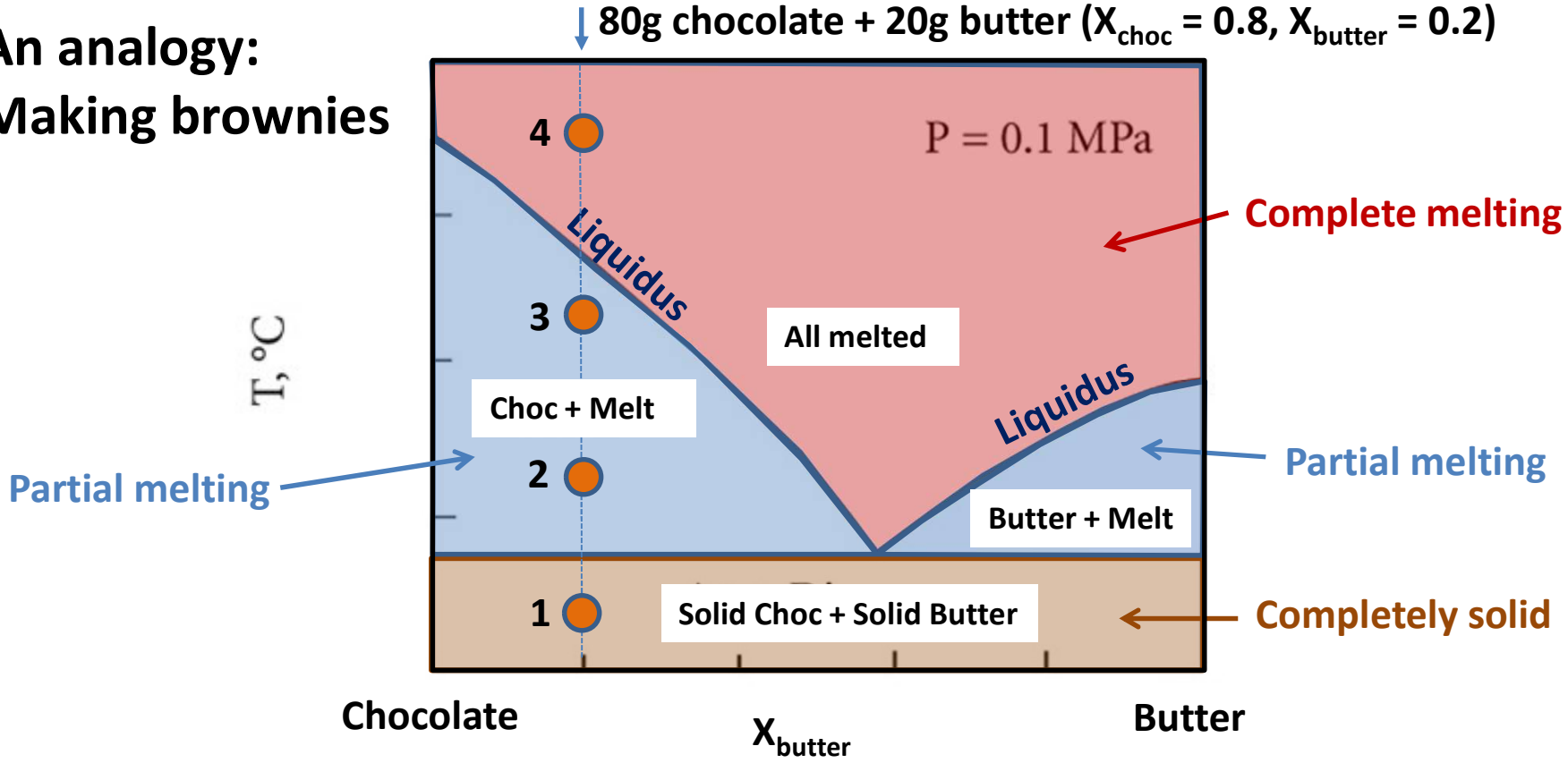
Diopside – melting point, 1391 °C.

PARTIAL MELTING:

A rock that consists of a mixture of minerals melts over a range of temperatures. When both melt and solid material are present, the rock is partially molten.



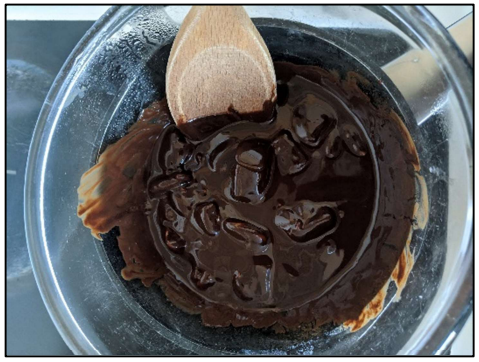
An analogy: Making brownies



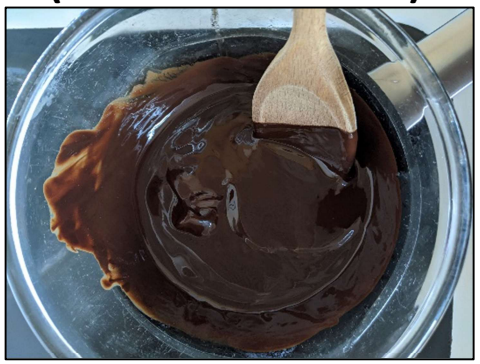
1 – Two solid phases



2 – Partial melt



3 – Partial melt (more melt than 2)



4 – Complete melt, one liquid



PARTIAL MELTING:

A rock that consists of a mixture of minerals melts over a range of temperatures. When both melt and solid material are present, the rock is partially molten.

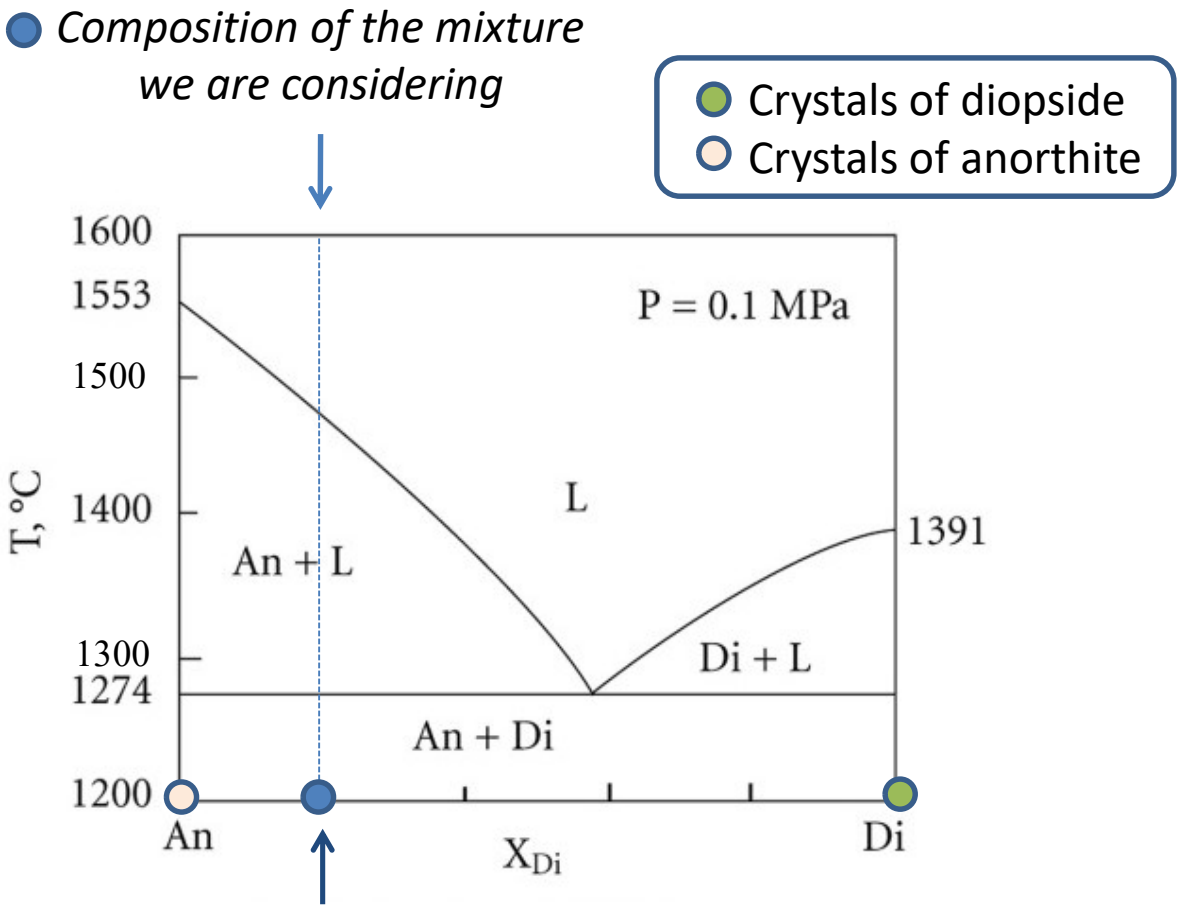
Example: A mixture of 80% Anorthite and 20% Diopside



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Gabbro:
plagioclase (anorthite)
and pyroxene (diopside)

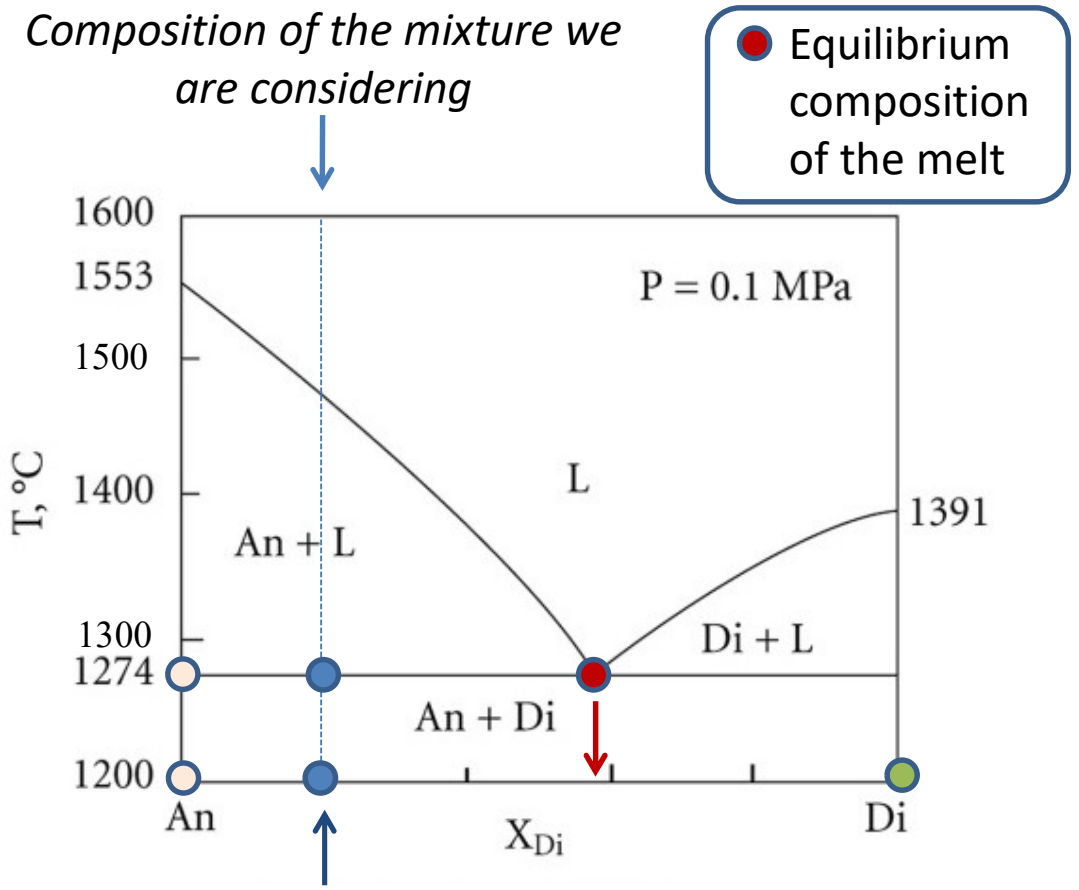
1200°C – two solid minerals



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Example: A mixture of 80% Anorthite and 20% Diopside



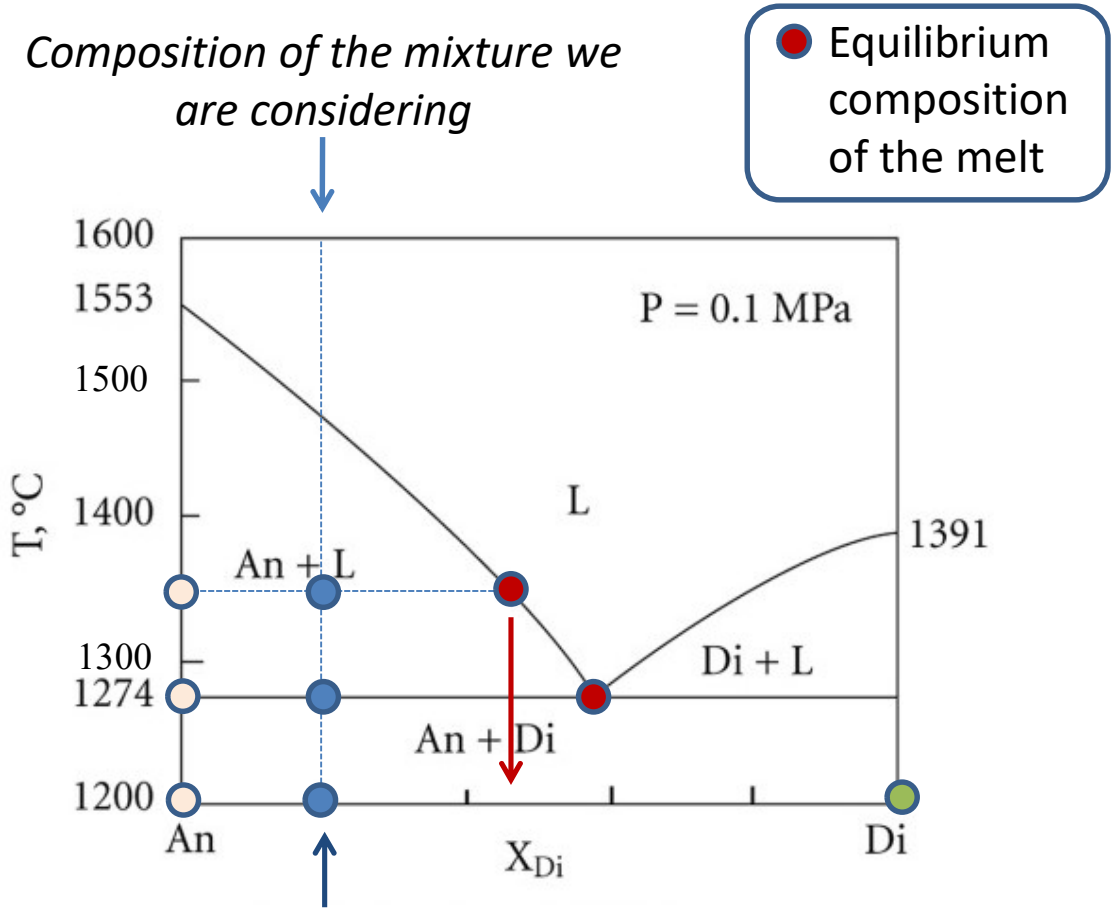
1274°C – first melting occurs

1200°C – two solid minerals

PARTIAL MELTING:

A rock that consists of a mixture of minerals melts over a range of temperatures. When both melt and solid material are present, the rock is partially molten.

Example: A mixture of 80% Anorthite and 20% Diopside



1350°C – partial melt

1274°C – first melting occurs

1200°C – two solid minerals

PARTIAL MELTING:

A rock that consists of a mixture of minerals melts over a range of temperatures. When both melt and solid material are present, the rock is partially molten.

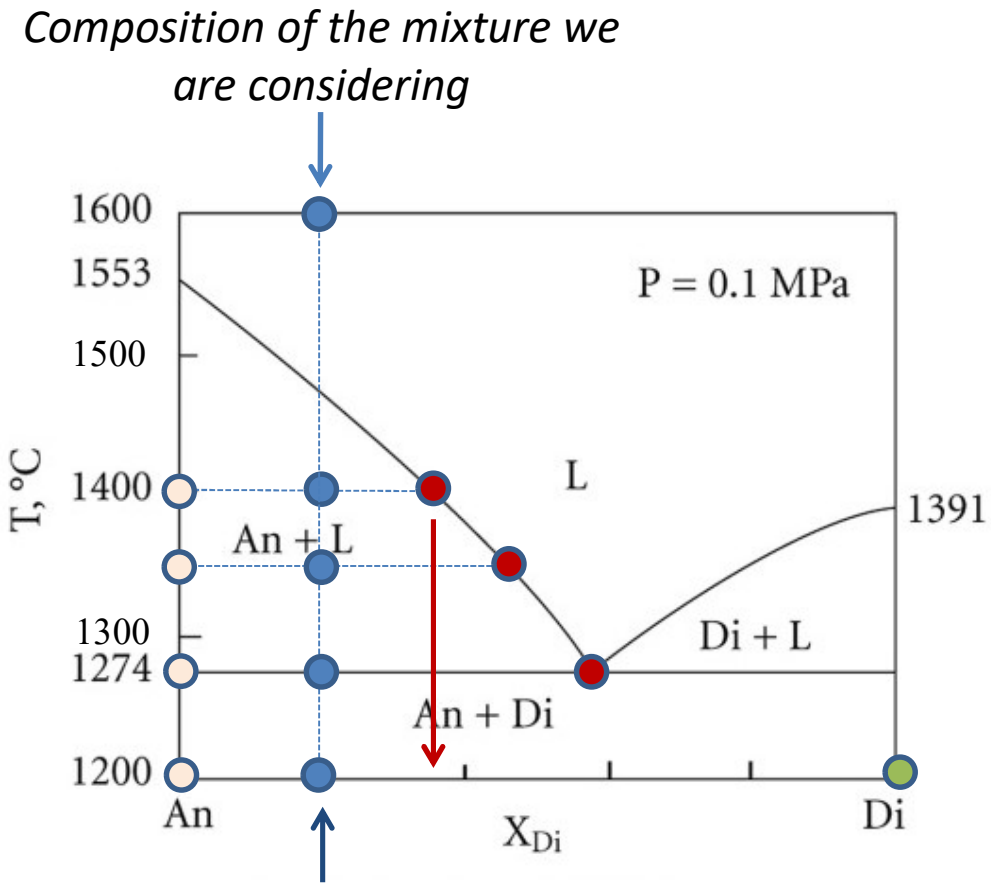
Example: A mixture of 80% Anorthite and 20% Diopside

1400°C – partial melt

1350°C – partial melt

1274°C – first melting occurs

1200°C – two solid minerals



Equilibrium composition of the melt

PARTIAL MELTING:

A rock that consists of a mixture of minerals melts over a range of temperatures. When both melt and solid material are present, the rock is partially molten.

Example: A mixture of 80% Anorthite and 20% Diopside

Composition of the mixture we are considering

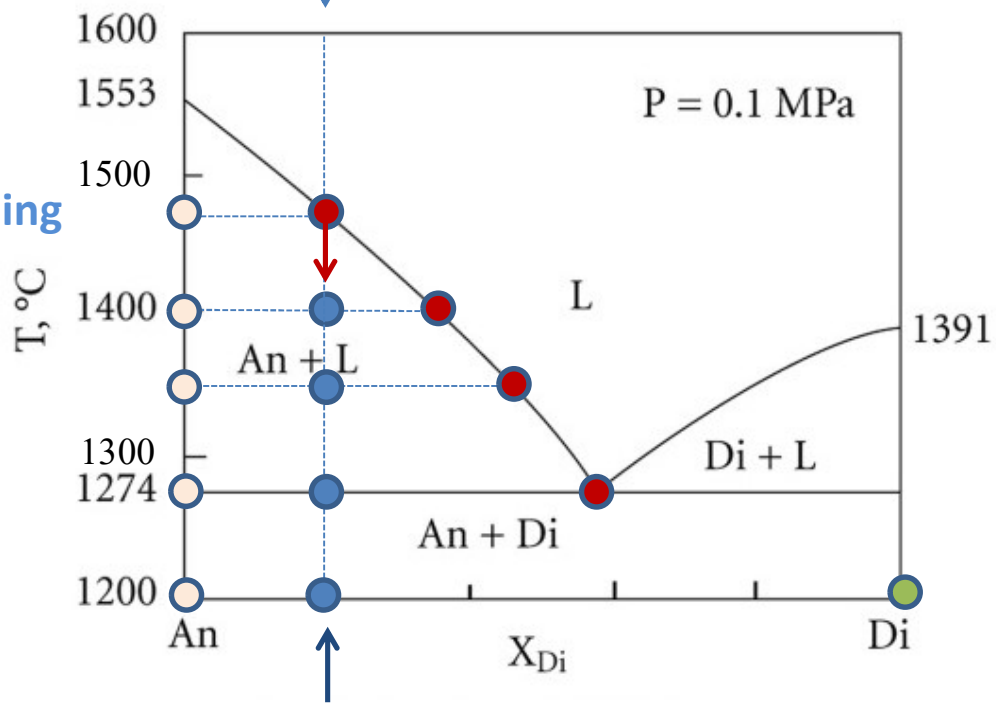
~1470°C – point of complete melting

1400°C – partial melt

1350°C – partial melt

1274°C – first melting occurs

1200°C – two solid minerals

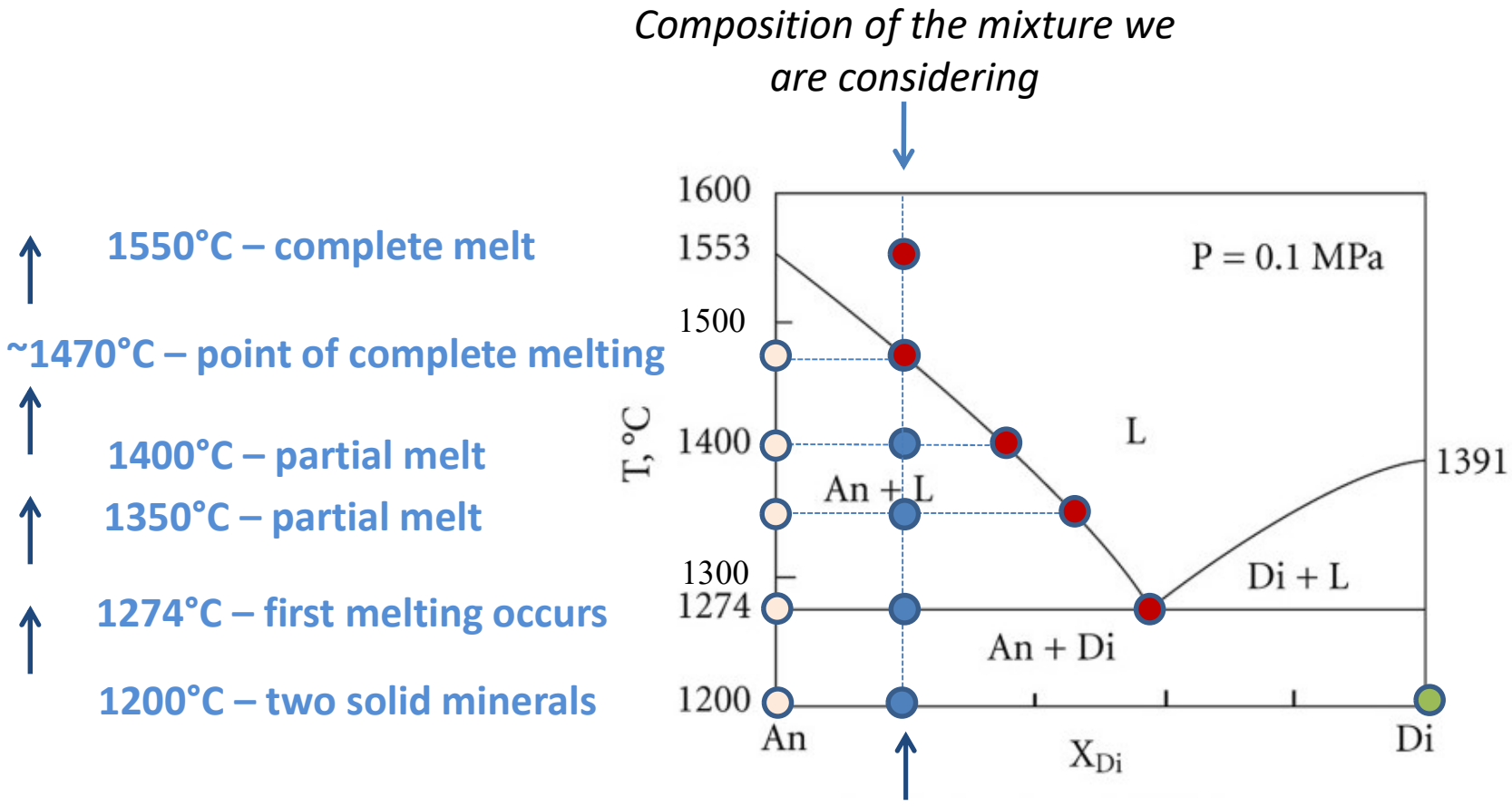


Equilibrium composition of the melt

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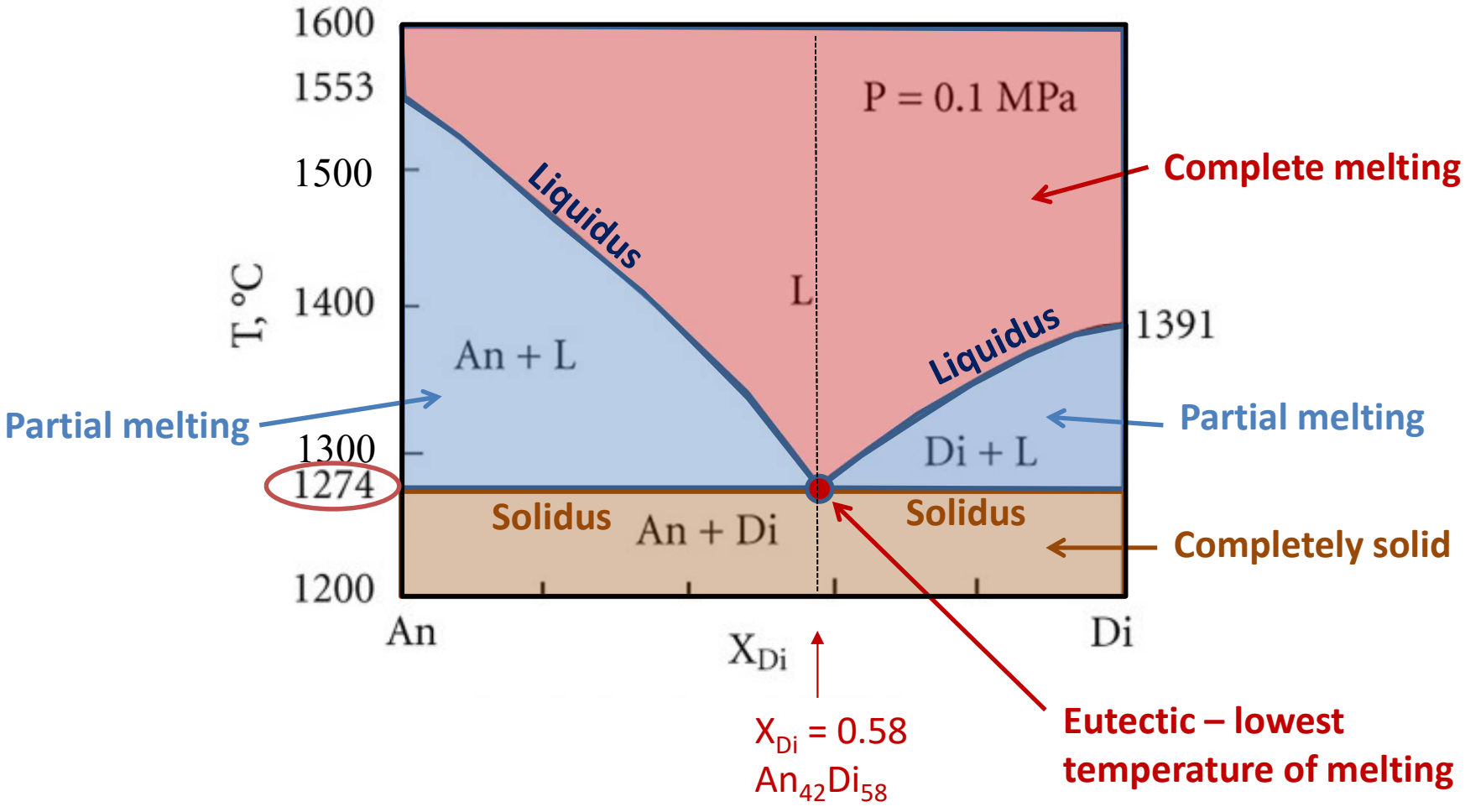
Example: A mixture of 80% Anorthite and 20% Diopside



Equilibrium composition of the melt

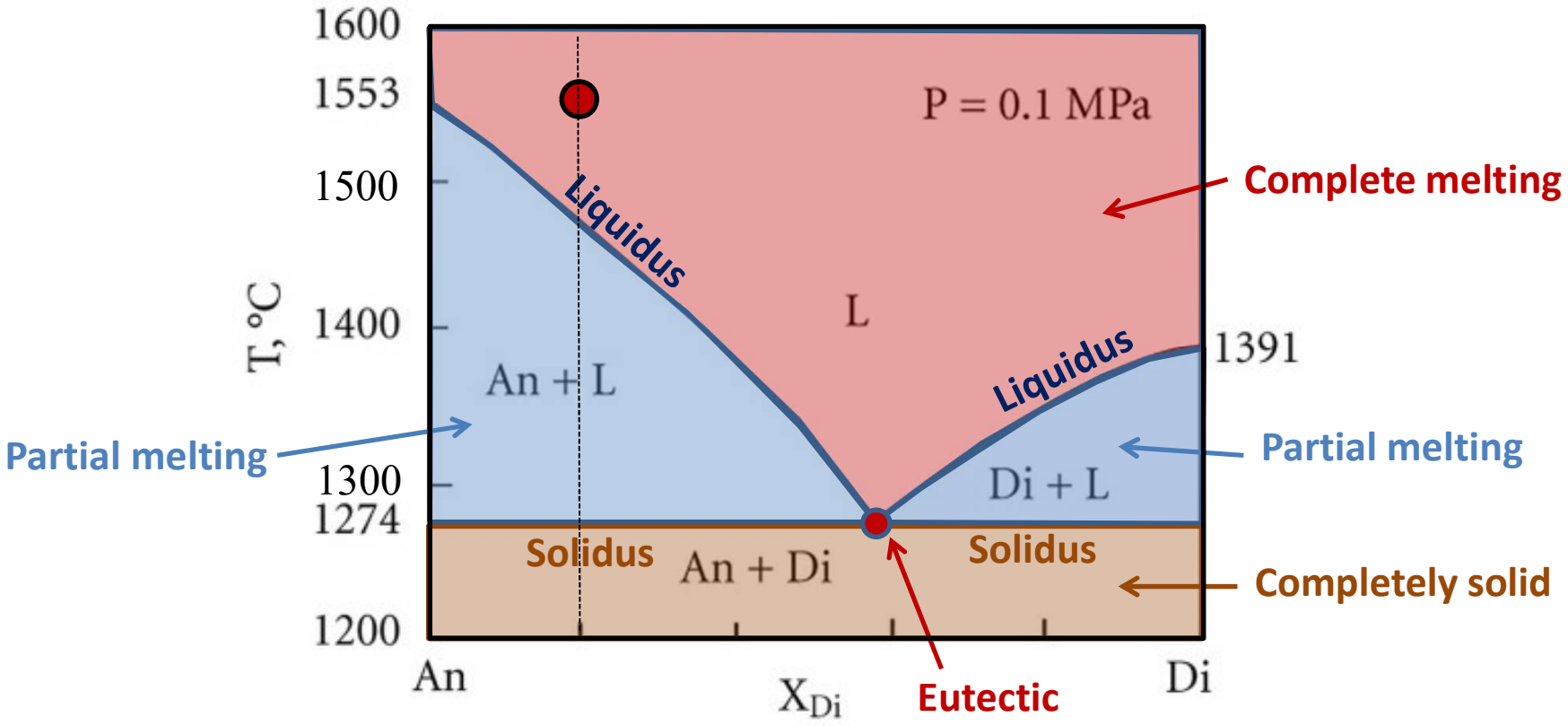
EUTECTIC MELTING

For any mixture of An & Di, the first melt appears at 1274 °C and has the composition of the eutectic ($An_{42}Di_{58}$)



SILICATE MELT (LIQUID) COMPOSITIONS

- Silicate melts are not ordered into crystalline structures.
- The silicate melt (liquid) can have any composition as long as it has charge balance – it is not restricted to a specific mineral formula.
- When we say a melt has a composition $An_{80}Di_{20}$ this means it has a composition equivalent to melting a mix containing 80% An and 20% Di crystals.

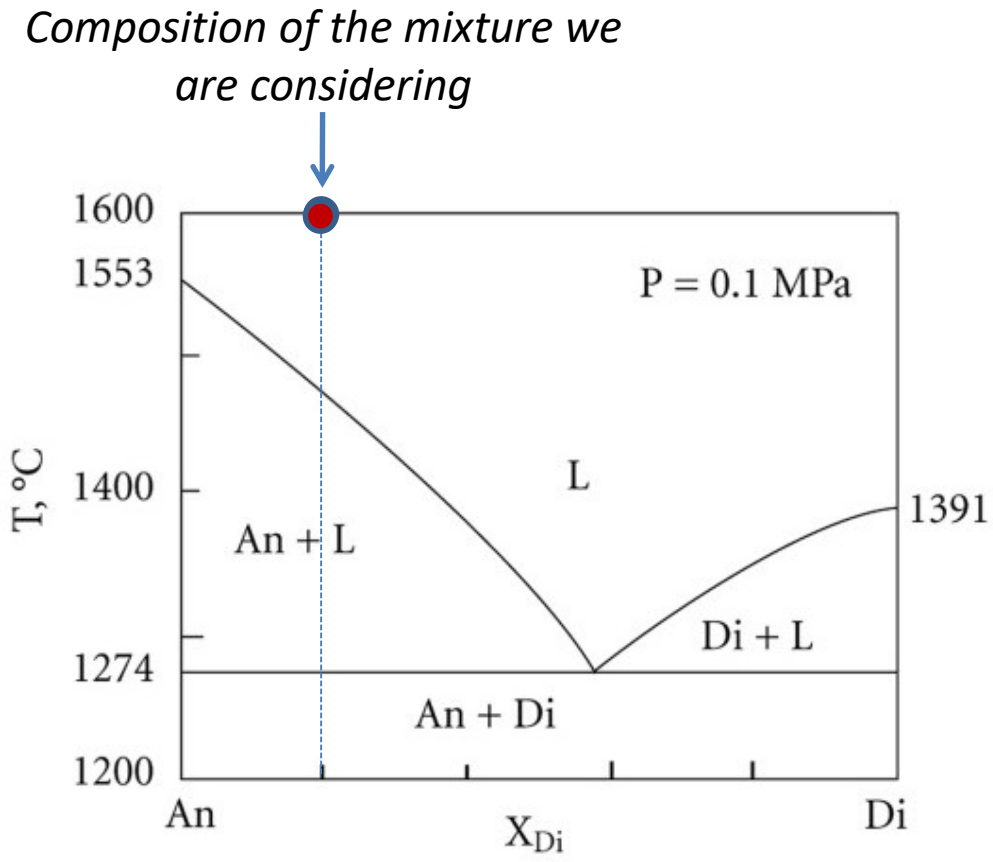


EQUILIBRIUM CRYSTALLIZATION – THE REVERSE OF MELTING

Equilibrium crystallization of a silicate melt follows the reverse path to melting.

Example: A mixture of 80% Anorthite and 20% Diopside

1600°C – complete melt



● Equilibrium composition of the melt

EQUILIBRIUM CRYSTALLIZATION – THE REVERSE OF MELTING

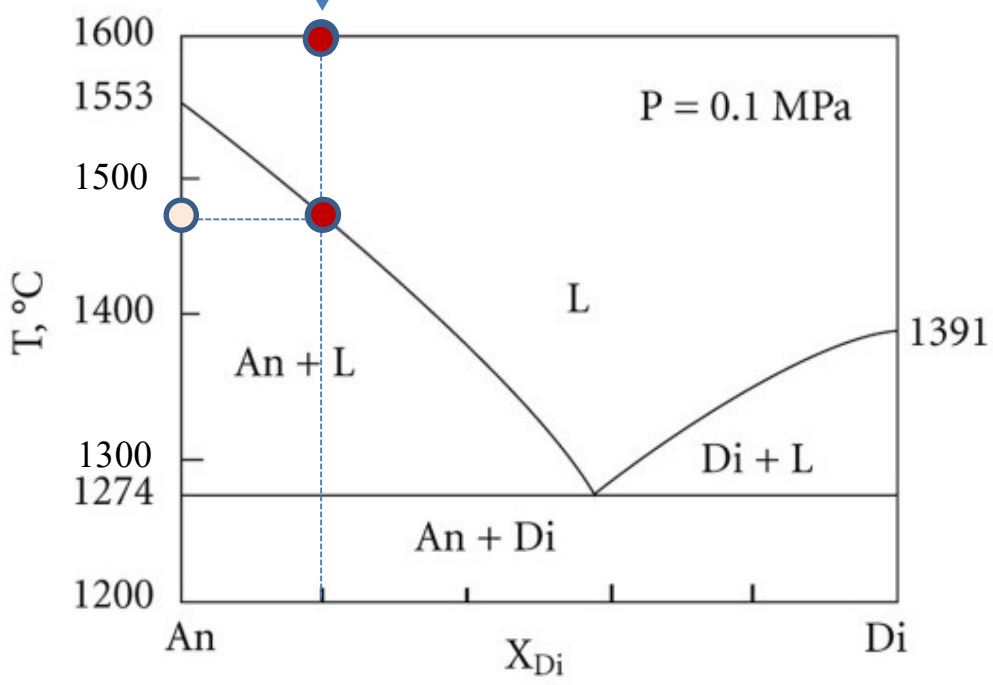
Equilibrium crystallization of a silicate melt follows the reverse path to melting.

Example: A mixture of 80% Anorthite and 20% Diopside

Composition of the mixture we are considering

1600°C – complete melt

~1470°C – point at which first solid appears



● Equilibrium composition of the melt

EQUILIBRIUM CRYSTALLIZATION – THE REVERSE OF MELTING

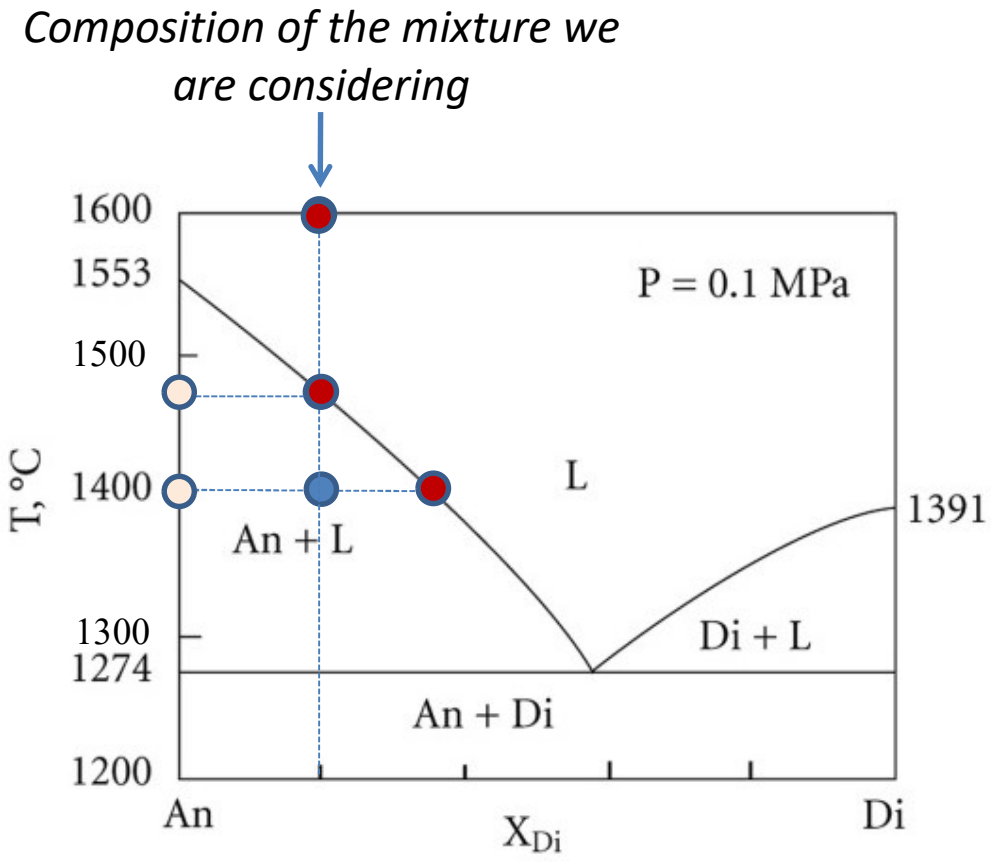
Equilibrium crystallization of a silicate melt follows the reverse path to melting.

Example: A mixture of 80% Anorthite and 20% Diopside

1600°C – complete melt

~1470°C – point at which first solid appears

1400°C – partial melt



Equilibrium composition of the melt

EQUILIBRIUM CRYSTALLIZATION – THE REVERSE OF MELTING

Equilibrium crystallization of a silicate melt follows the reverse path to melting.

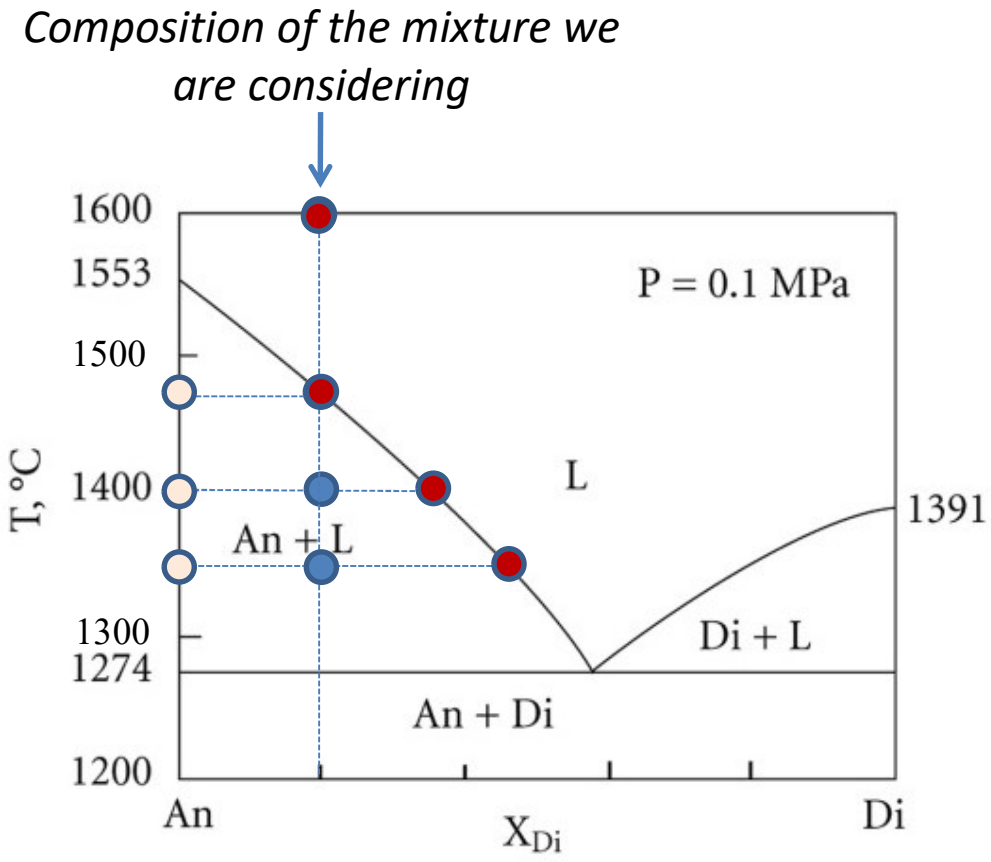
Example: A mixture of 80% Anorthite and 20% Diopside

1600°C – complete melt

~1470°C – point at which first solid appears

1400°C – partial melt

1350°C – partial melt



Equilibrium composition of the melt

EQUILIBRIUM CRYSTALLIZATION – THE REVERSE OF MELTING

Equilibrium crystallization of a silicate melt follows the reverse path to melting.

Example: A mixture of 80% Anorthite and 20% Diopside

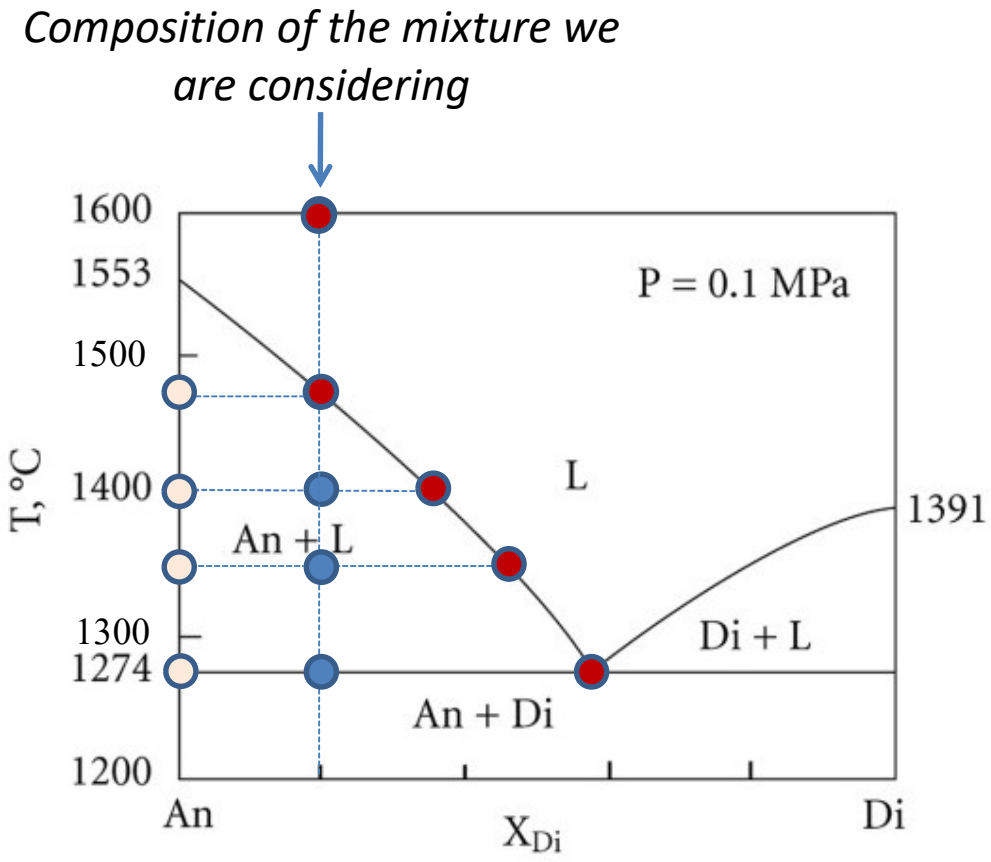
1600°C – complete melt

~1470°C – point at which first solid appears

1400°C – partial melt

1350°C – partial melt

1274°C – last melt present



Equilibrium composition of the melt

EQUILIBRIUM CRYSTALLIZATION – THE REVERSE OF MELTING

Equilibrium crystallization of a silicate melt follows the reverse path to melting.

Example: A mixture of 80% Anorthite and 20% Diopside



Gabbro

1600°C – complete melt

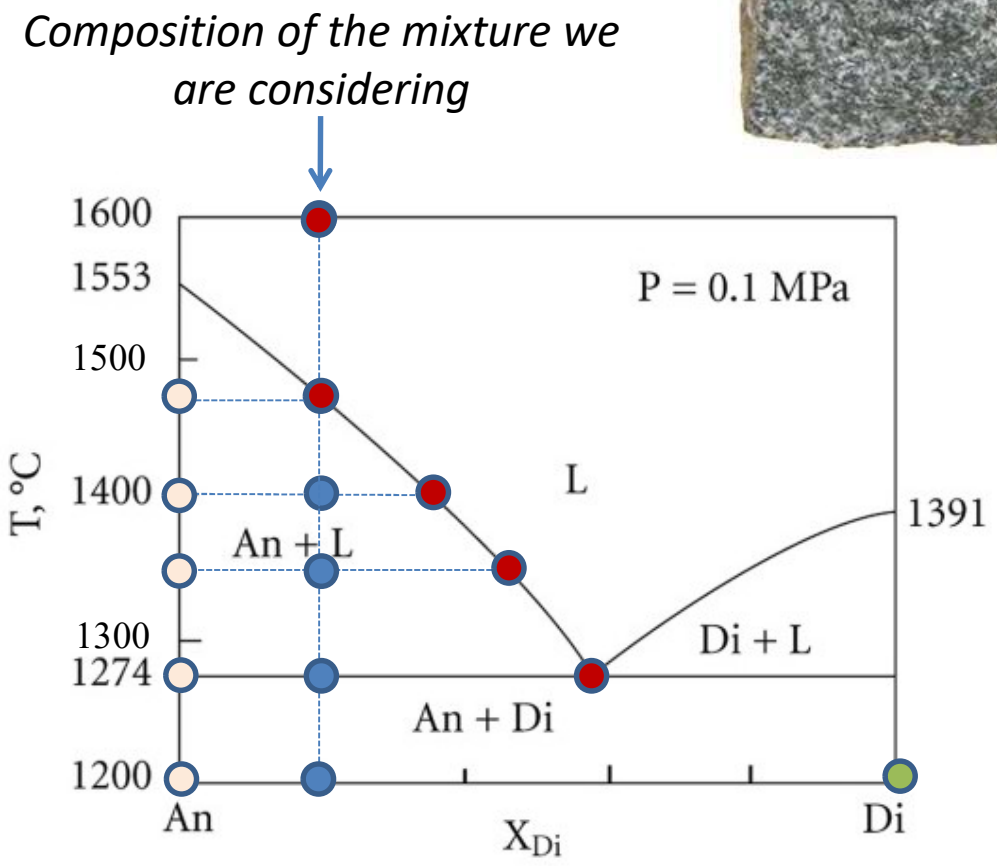
~1470°C – point at which first solid appears

1400°C – partial melt

1350°C – partial melt

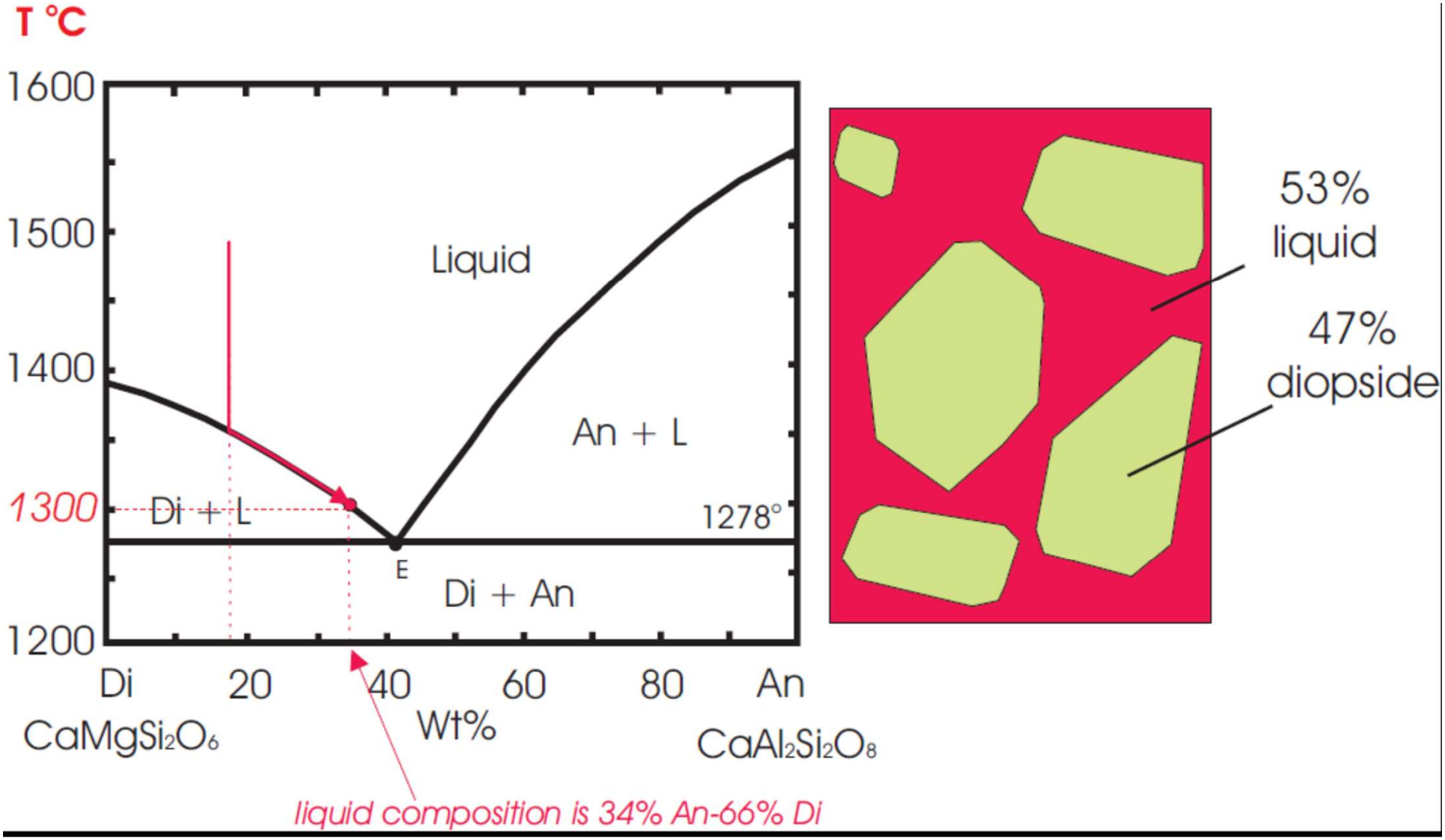
1274°C – last melt present

1200°C – two solid minerals



Equilibrium composition of the melt

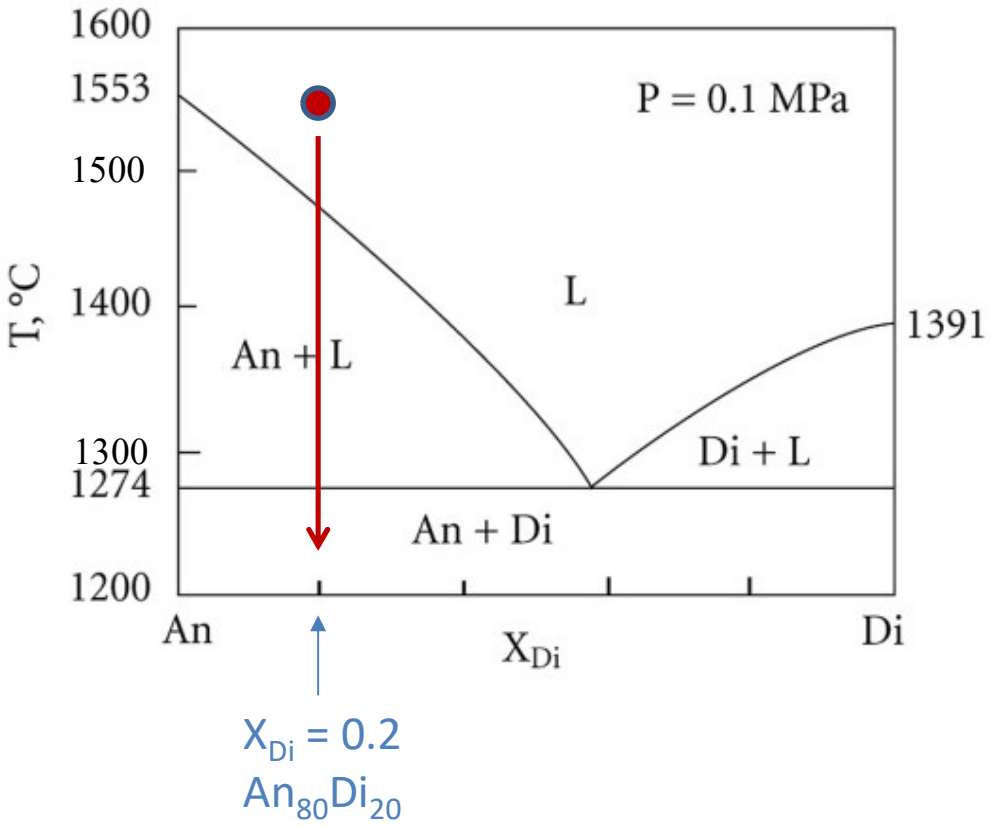
See: Diopside-Anorthite.pdf for a complete crystallisation path



Red line shows how the composition of the liquid changes as the system cools

How to read a binary (two-component) phase diagram

Example: Bulk composition $An_{80}Di_{20}$



1550 °C

Composition of interest is in a one-phase region of the diagram (L, liquid).

At 1500 °C, the system is completely melted. Only one phase is stable – liquid. To find the composition of the liquid, drop a line vertically and read the composition off the x axis.

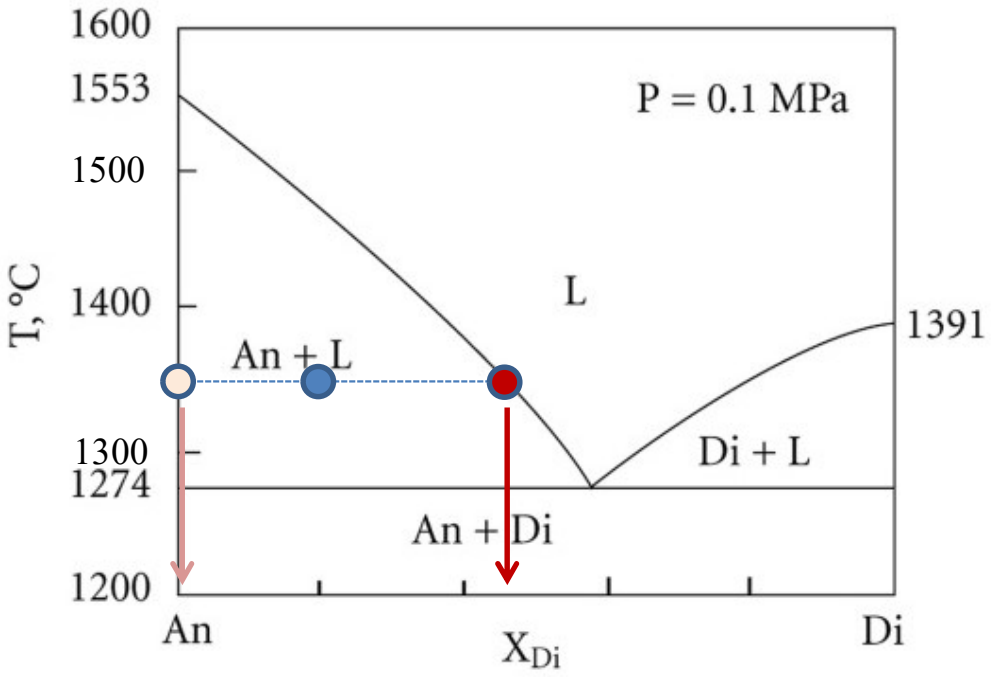
Composition of the liquid is

$An_{80}Di_{20}$

- Not surprising since this is the composition of the system.

How to read a binary (two-component) phase diagram

Example: Bulk composition $An_{20}Di_{80}$



X_{Di} = 0.45
$An_{55}Di_{45}$

1350 °C

Composition of interest is in a two-phase region of the diagram.

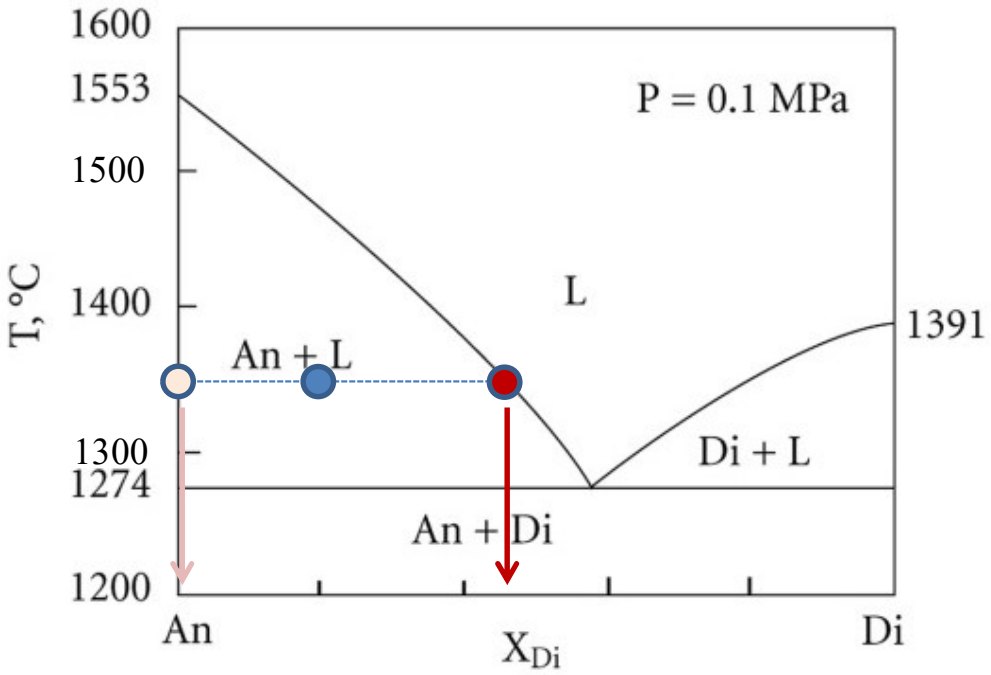
At 1350 °C, the system consists of anorthite plus liquid.

The composition of the **solid** is 100% anorthite (**An_{100}**).

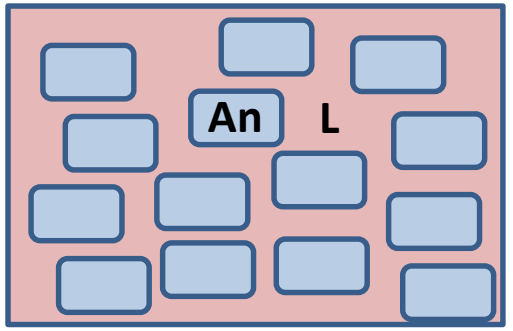
To find the *composition* of the liquid, drop a vertical line down to the x axis. Composition of the **liquid** is about **$An_{55}Di_{45}$**.

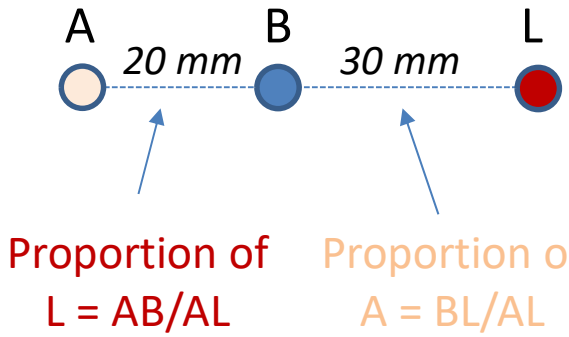
How to read a binary (two-component) phase diagram

Example: Bulk composition $An_{20}Di_{80}$



The diagram also tells us *how much* liquid there is, i.e. the *proportions* of anorthite and liquid in the system.





Lever Rule:

The proportion of anorthite is given by: BL / AL

The proportion of liquid is given by: AB / AL

Example:

Proportion of liquid = 20/50 = 40%

40% of the system is liquid, 60% is solid anorthite

Brownies – finished product



Melting chocolate and butter is an analogy for melting.

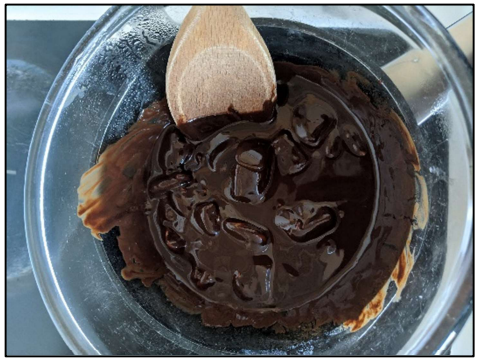
BUT the analogy breaks down on cooling:

- The chocolate-butter liquid would not cool down to the situation in (1).
- For a rock, cooling is exactly the reverse of heating, and the rock will crystallize to a state with the two solid minerals analogous to (1).

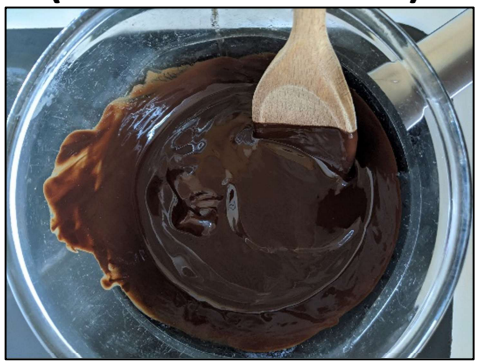
1 – Two solid phases



2 – Partial melt



3 – Partial melt (more melt than 2)



4 – Complete melt, one liquid

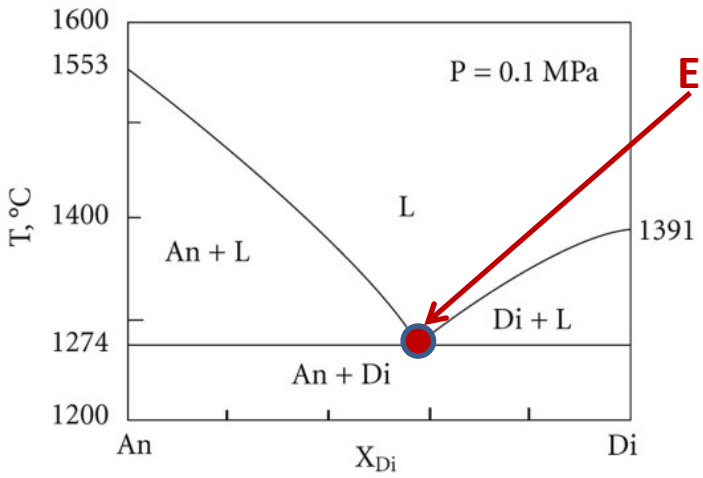


Chemistry of the Earth: Lecture 7
Melting and the Rare Earth Elements (REE)



Part 2: Melting the mantle and the REE

EUTECTIC MELTING



Eutectic



Gabbro: plagioclase and pyroxene

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For any mixture of An & Di, the first melt appears at 1274 °C and has the composition of the eutectic ($An_{42}Di_{58}$)

THE REAL WORLD

Most rocks contain more than two minerals. However, the same principles apply:

- The first melt appears at the eutectic temperature
- For a given set of minerals, the melt has the same composition irrespective of the proportions of the minerals.

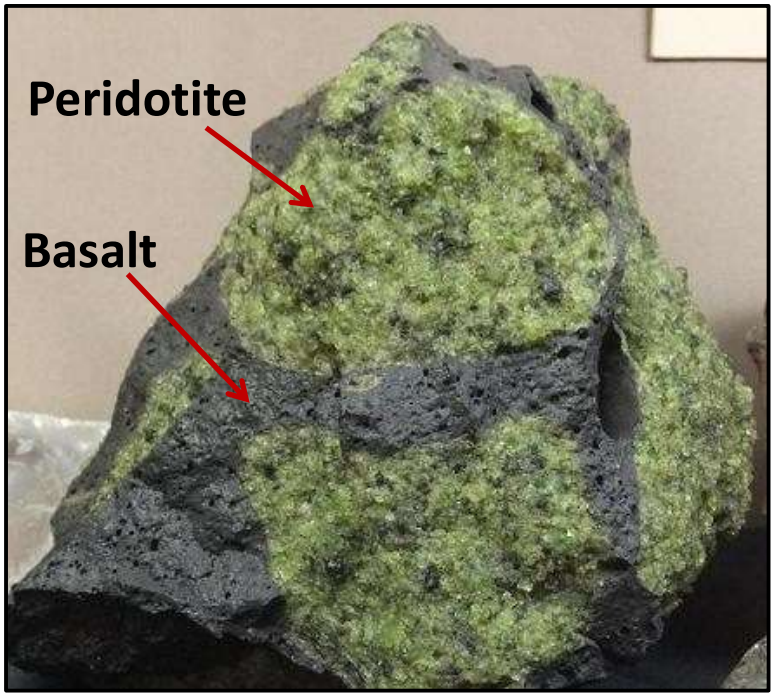


Granite: plagioclase, alkali feldspar, quartz, mica

MELTING THE MANTLE

In a very simplistic view of the Earth, when the mantle (which consists of peridotite) begins to melt, a eutectic melt is produced that has the composition of a basalt.

LECTURE 1



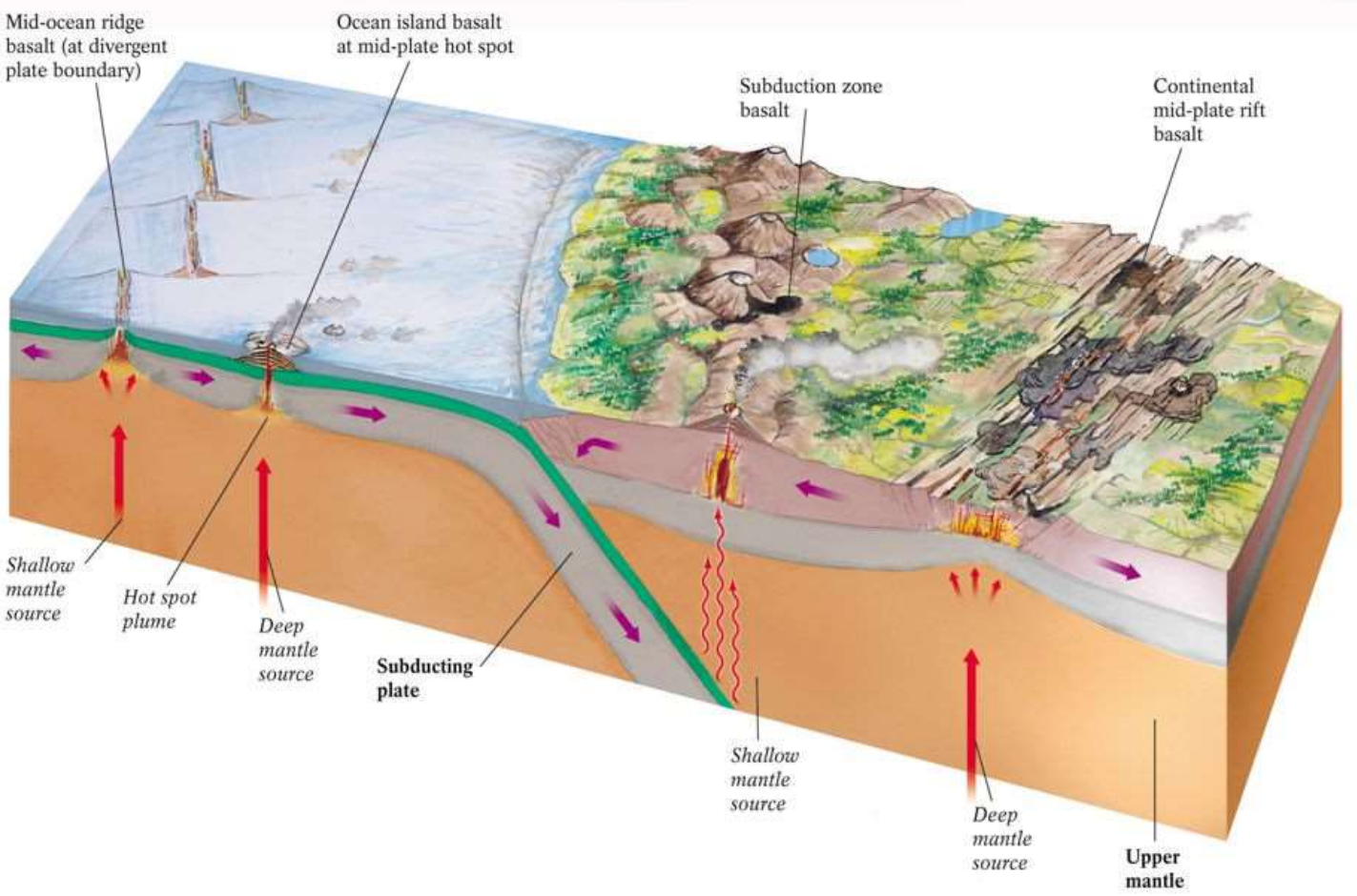
http://peridot1.blogspot.co.uk/

The melt is buoyant and rises through the mantle and crust, then erupts as a basalt.



http://hvo.wr.usgs.gov/archive/2004_07_28.html

MELTING THE MANTLE: PRODUCTION OF BASALT

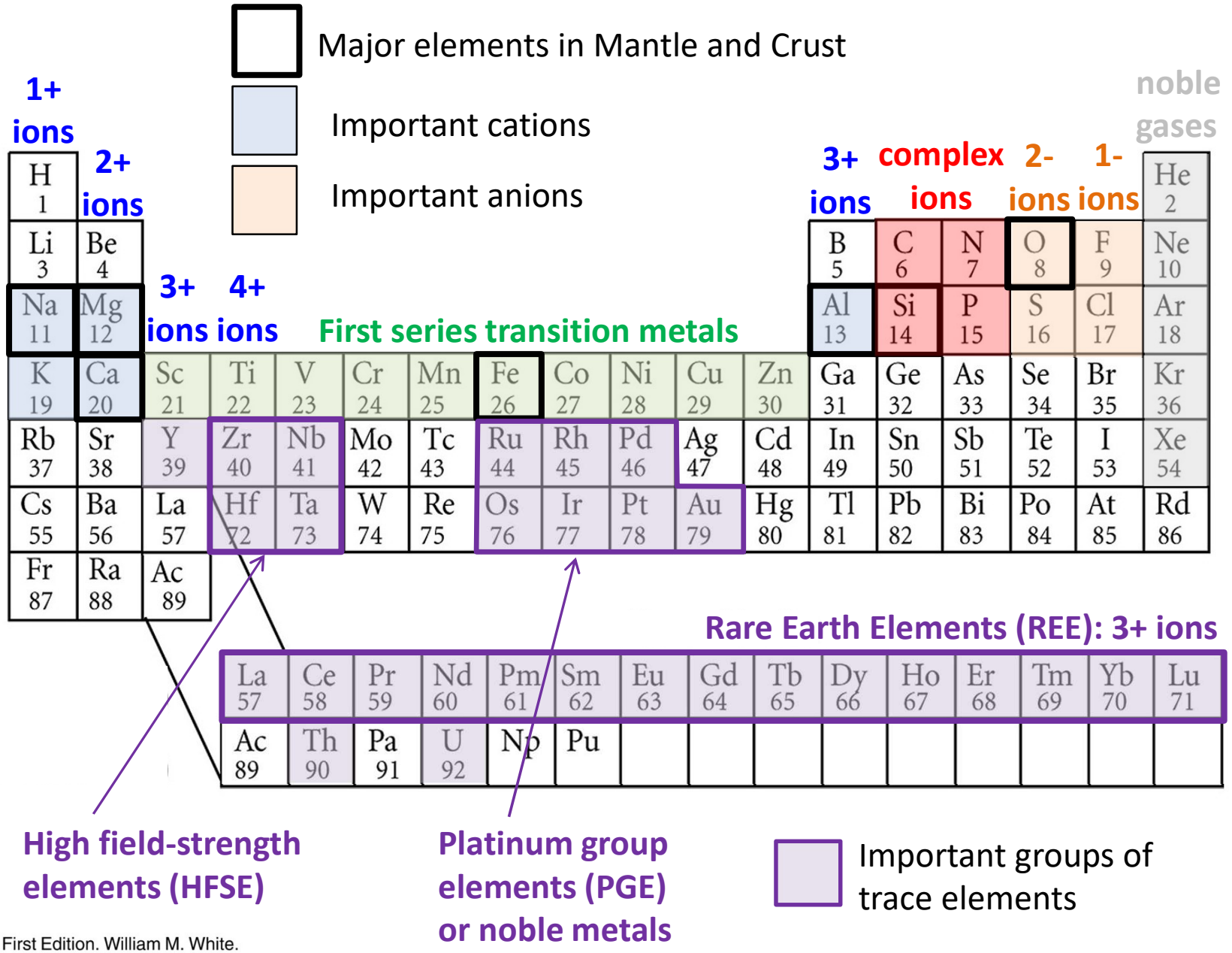


- Basalts form in many different plate tectonic settings.
- Basalts form by melting of the mantle at different depths.
- We will look at how the Rare Earth Elements (REE) are powerful trace elements that help understand melting processes.

http://slideplayer.com/slide/6010814/

A GEOCHEMIST'S VIEW OF THE PERIODIC TABLE

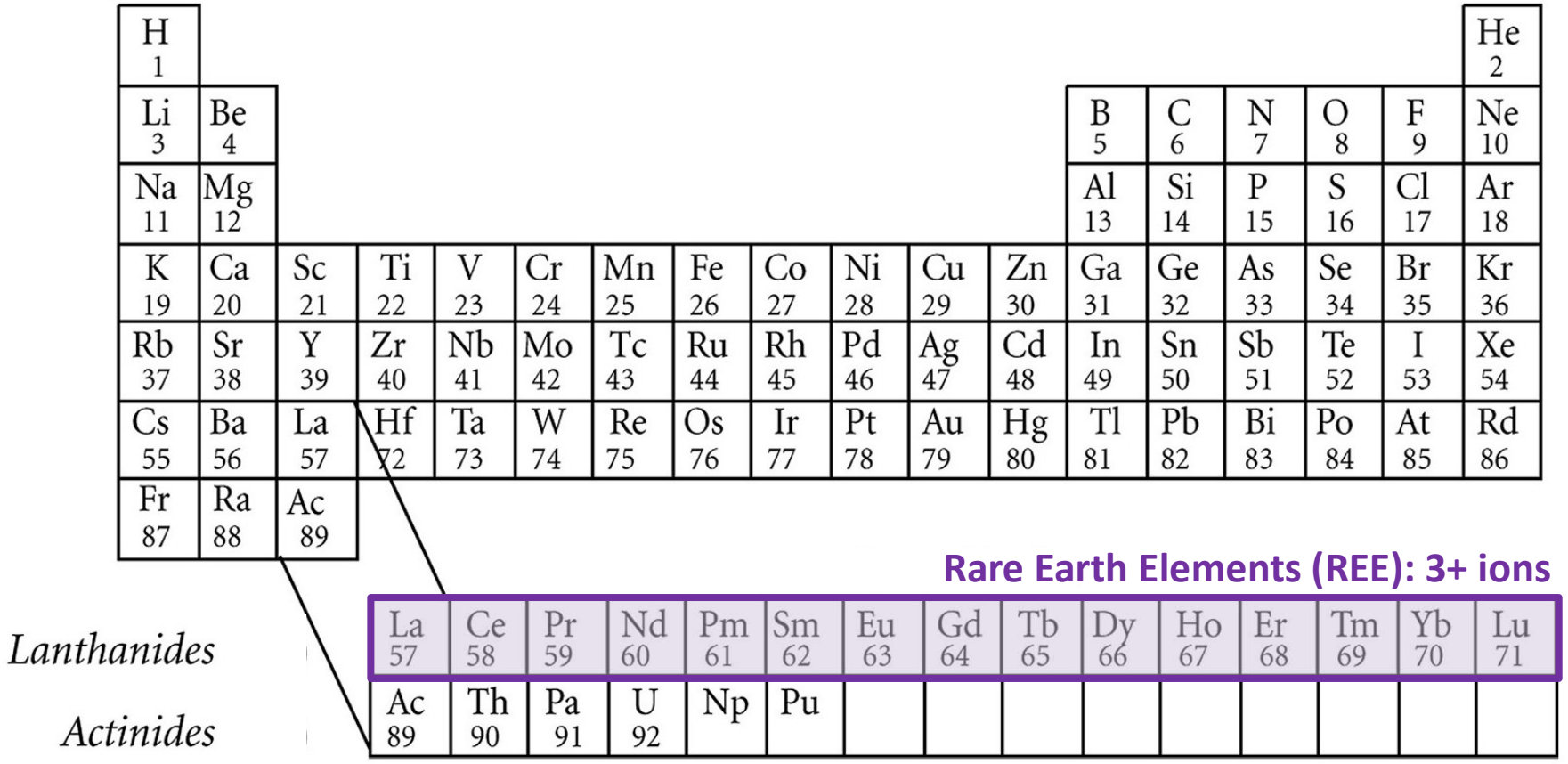
LECTURE 2



Geochemistry, First Edition. William M. White.
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THE RARE EARTH ELEMENTS (REE)

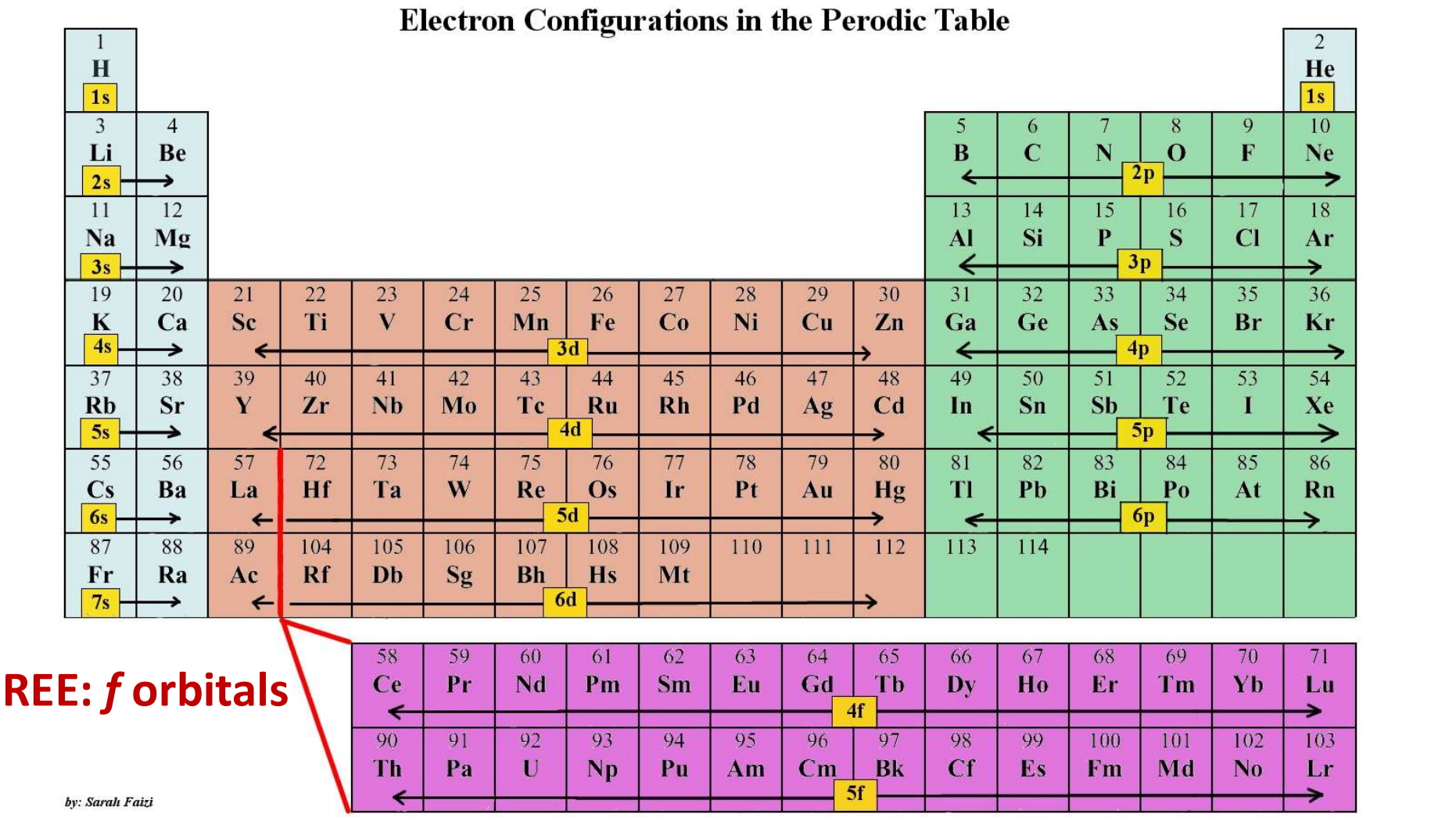
- also known as LANTHANIDES, first element is Lanthanum (La)



Geochemistry, First Edition. William M. White.
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The arrangement of elements in the Periodic Table follows the electron configurations. Yellow boxes e.g. 2p below refer to the orbitals that are being filled. Orbitals with lower energy levels are assumed to be full.

LECTURE 2

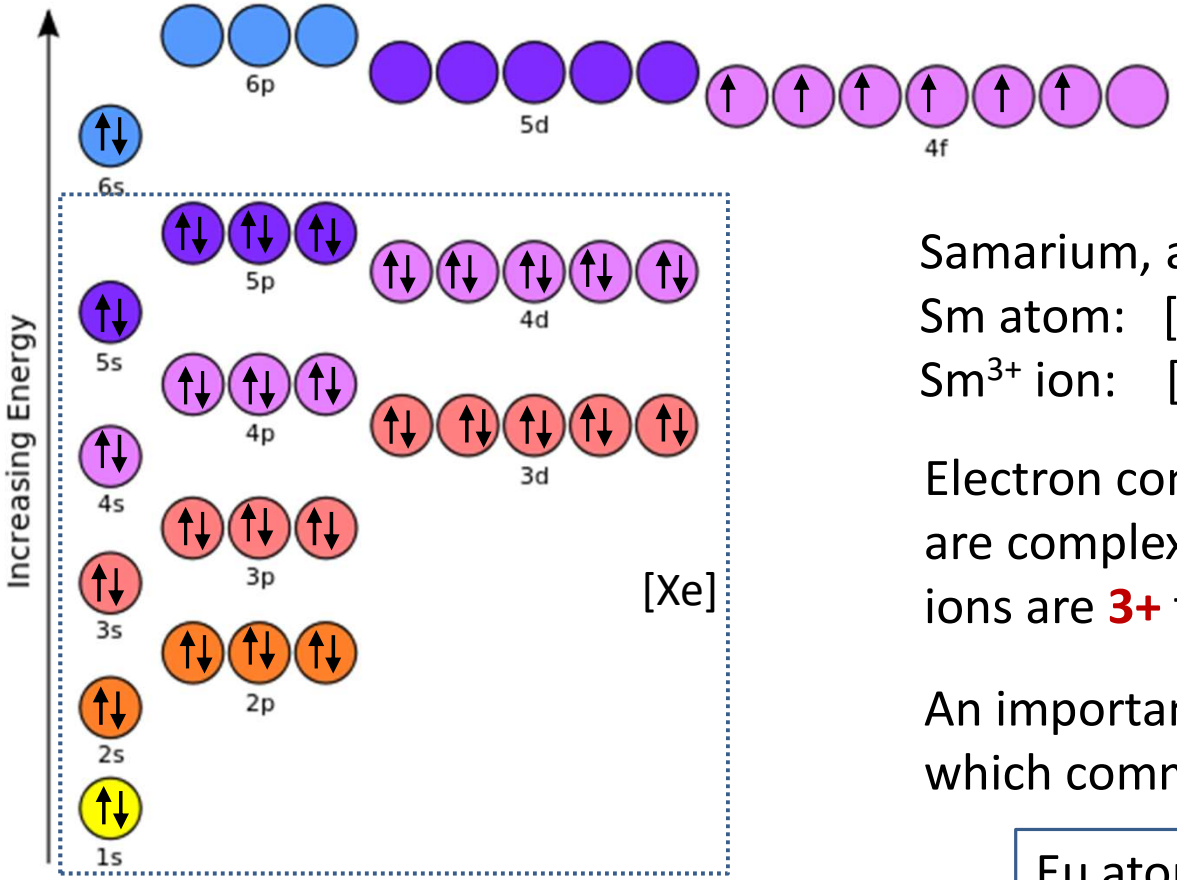


REE: *f* orbitals

http://chemwiki.ucdavis.edu/Core/Inorganic_Chemistry/Electronic_Structure_of_Atoms_and_Molecules/Electronic_Configurations

REE: Electron configurations

LECTURE 2



Samarium, atomic number 62

Sm atom: [Xe] $4f^6\ 6s^2$

Sm^{3+} ion: [Xe] $4f^5$

Electron configurations for the REE are complex, but the most stable ions are **3+** for all the REE.

An important exception is Eu (Z = 63) which commonly occurs as **Eu^{2+}**:

Eu atom: [Xe] $4f^7\ 6s^2$

Eu^{2+} ion: [Xe] $4f^7$

Eu^{3+} ion: [Xe] $4f^6$

Also, Ce occurs as Ce^{4+}

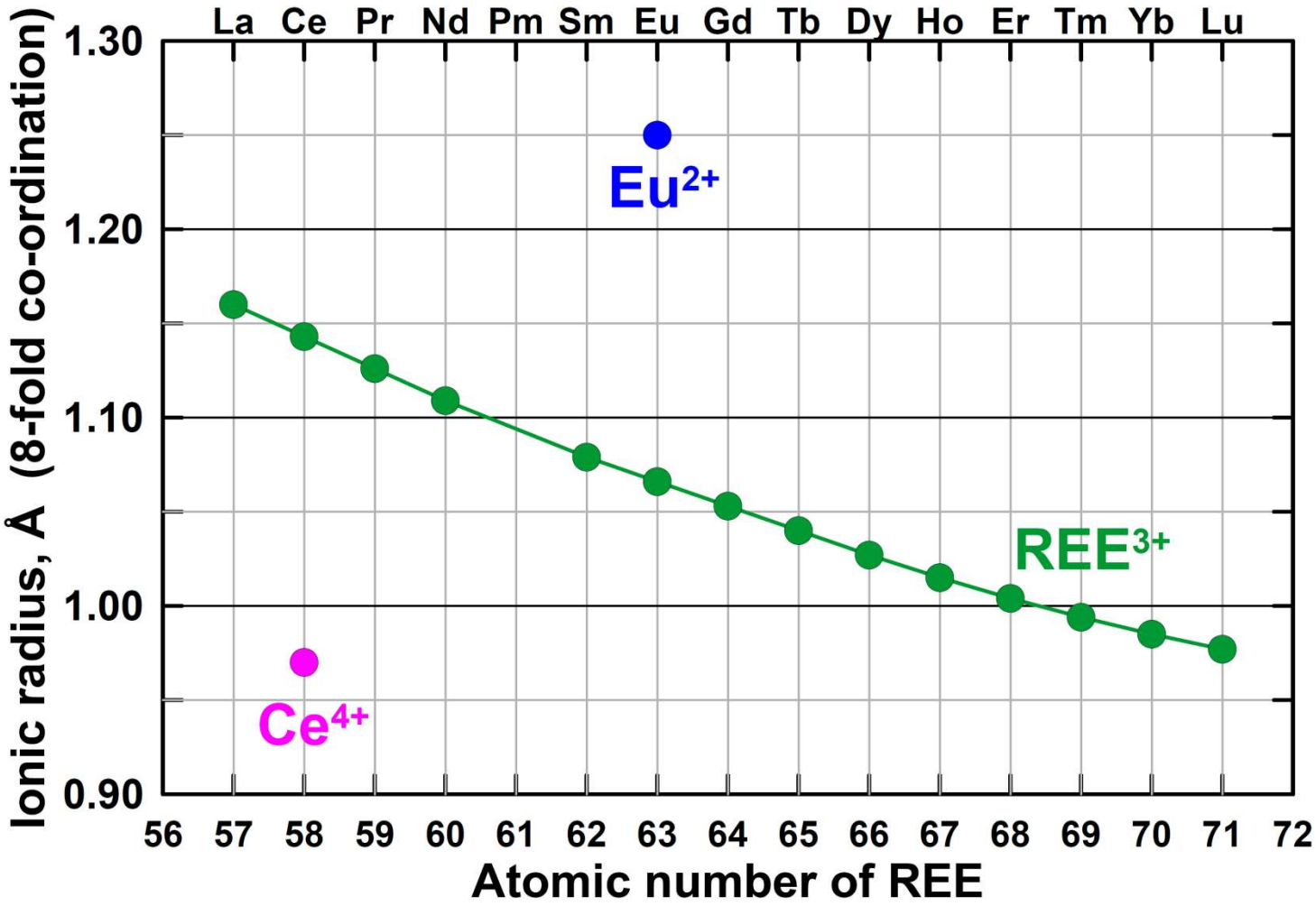
REE: Ionic radii for 3+ ions

Atomic number	Name	Symbol	Ionic radius for eight-fold coordination *	
57	Lanthanum	La	La^{3+}	1.160
58	Cerium	Ce	Ce^{3+}	1.143
			Ce^{4+}	0.970
59	Praesodymium	Pr	Pr^{3+}	1.126
60	Neodymium	Nd	Nd^{3+}	1.109
61	Promethium	Pm	Not naturally occurring	
62	Samarium	Sm	Sm^{3+}	1.079
63	Europium	Eu	Eu^{3+}	1.066
			Eu^{2+}	1.250
64	Gadolinium	Gd	Gd^{3+}	1.053
65	Terbium	Tb	Tb^{3+}	1.040
66	Dysprosium	Dy	Dy^{3+}	1.027
67	Holmium	Ho	Ho^{3+}	1.015
68	Erbium	Er	Er^{3+}	1.004
69	Thulium	Tm	Tm^{3+}	0.994
70	Ytterbium	Yb	Yb^{3+}	0.985
71	Lutetium	Lu	Lu^{3+}	0.977
39	Yttrium	Y	Y^{3+}	1.019

Ionic radius decreases

* From Shannon (1976), in Ångstroms (10^{-10} m).

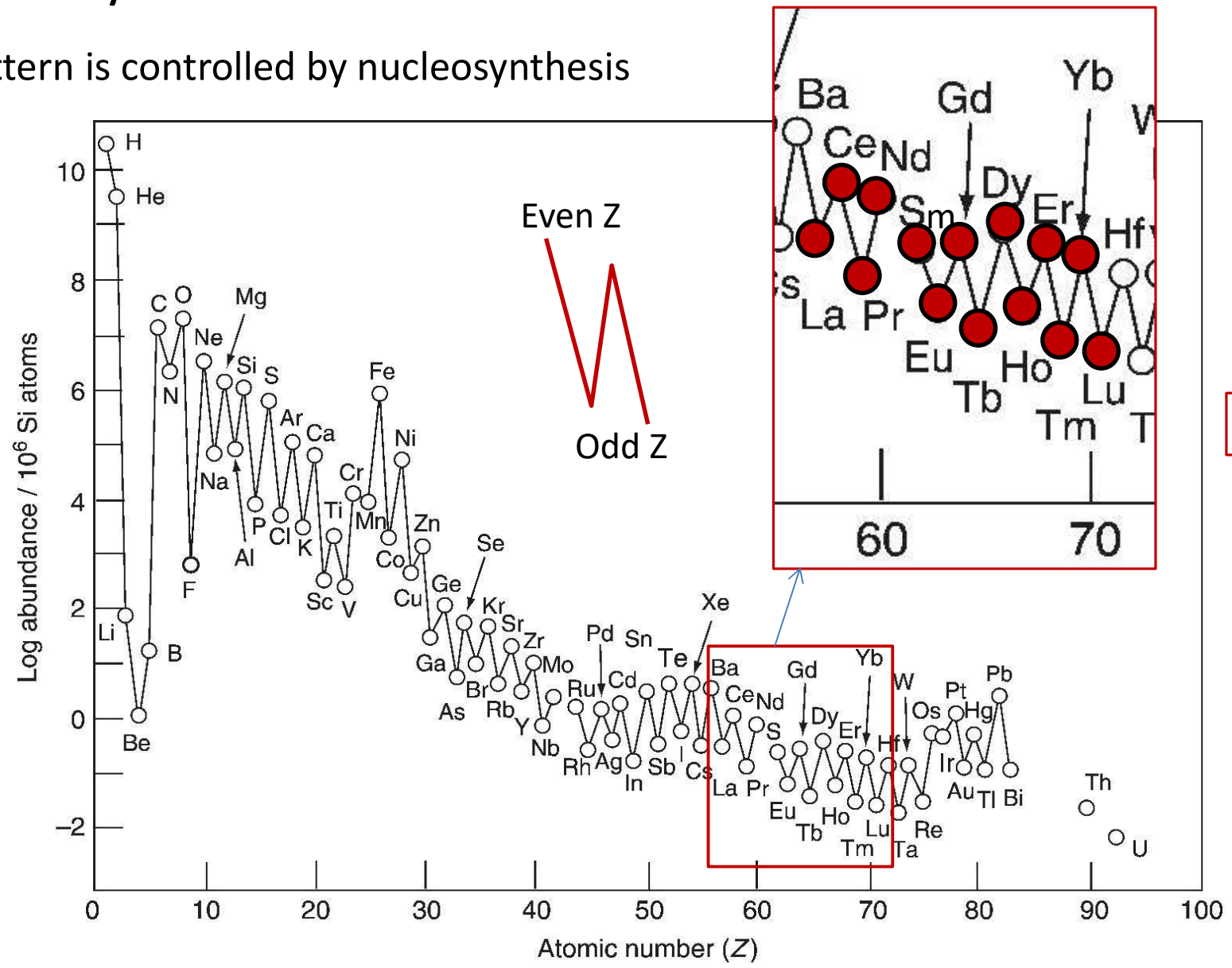
REE: Lanthanide contraction



- The ionic radius of REE ions with charge 3+ decreases steadily with increasing Z.
- Reason is that the *f* orbital electrons do not completely shield the charge of the nucleus. As the charge on the nucleus increases, the electrons are pulled closer in.

REE: Odd / even abundances in nature

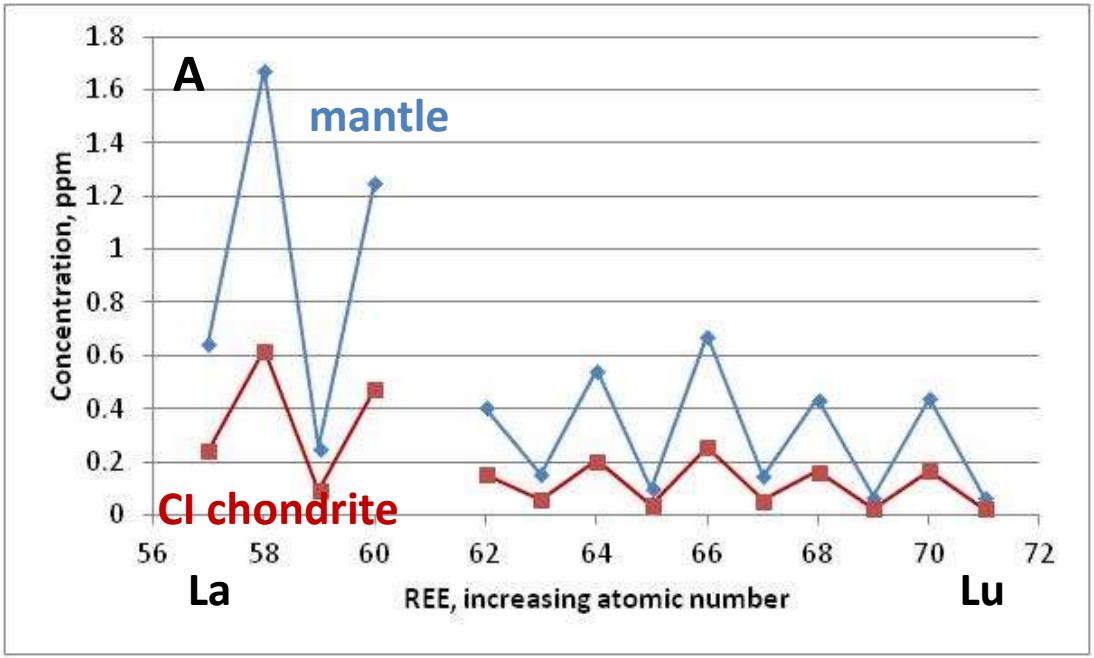
Pattern is controlled by nucleosynthesis



LECTURE 1

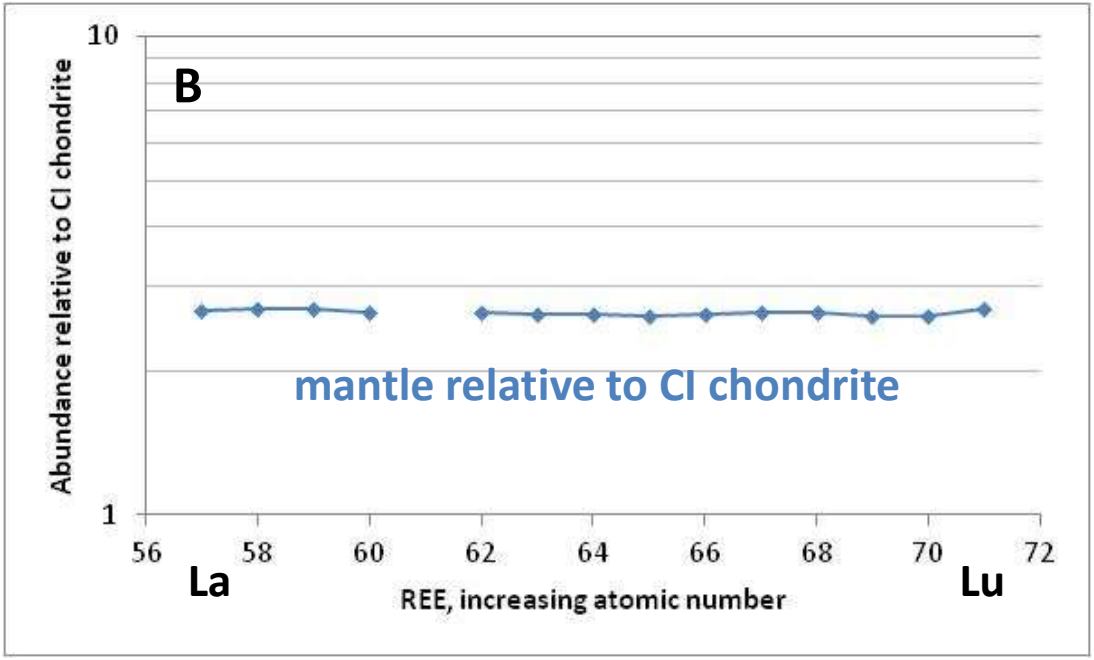
Introduction to Geochemistry: Principles and Applications, First Edition. Kula C. Misra.
© 2012 Kula C. Misra. Published 2012 by Blackwell Publishing Ltd.

REE: “Normalizing” to remove the odd/even pattern



A. The actual concentrations of REE in the Earth’s mantle or in CI chondrites show the zig-zag pattern inherited from nucleosynthesis.

{Promethium (Pm), Z = 61, is not naturally occurring}



B. Dividing the concentrations in the mantle by the concentrations in CI chondrites smooths out the pattern. This trick is called “normalization”.

{REE abundances are commonly plotted on a logarithmic scale.}

PARTITION COEFFICIENTS BETWEEN MINERALS AND MELTS

We have previously looked at partitioning between two liquids (lecture 3).
Now we will consider how elements partition between minerals and melts.

Mineral / melt partition coefficient:

$$\text{Partition coefficient (D)} = \frac{\text{Concentration in mineral}}{\text{Concentration in melt}}$$

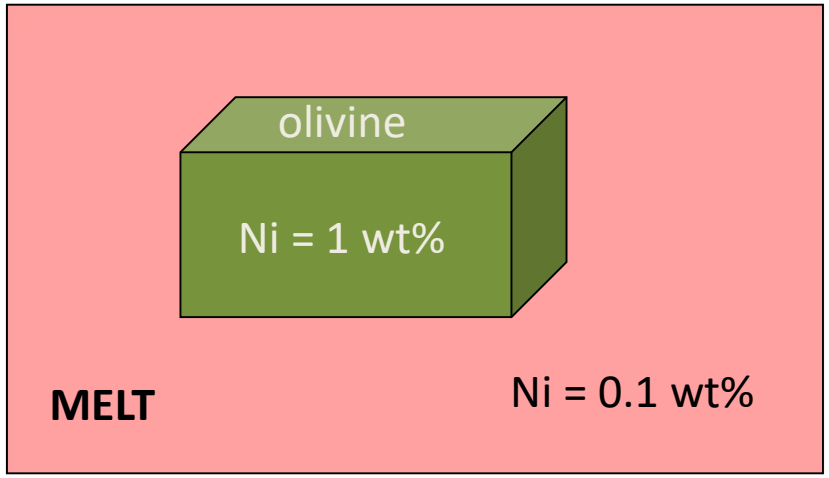
Concentrations are measured as weight percent (wt%) or ppm values.

Example:

Olivine / melt partition coefficient for nickel $^{Ni}D_{olv/melt} = \frac{\text{wt\% Ni in olivine}}{\text{wt\% Ni in melt}}$

$= 1 / 0.1$

$= 10$



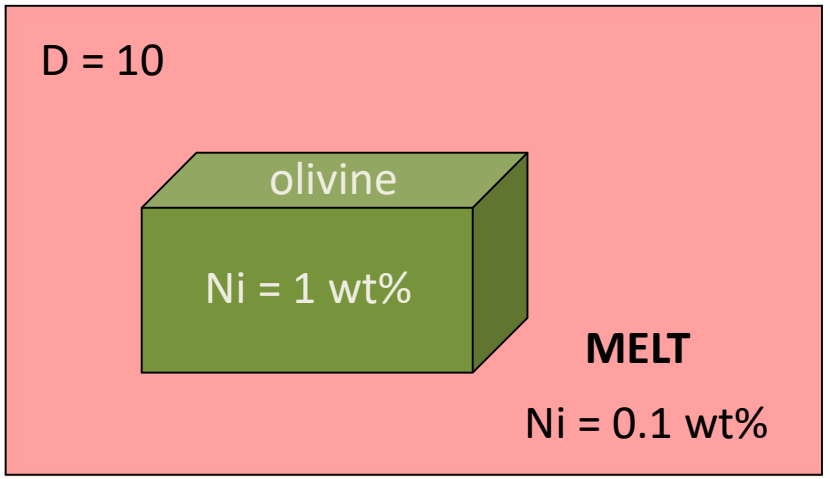
Concentration of Ni is 10 times higher in olivine than in melt

{D is a ratio so it has no units}

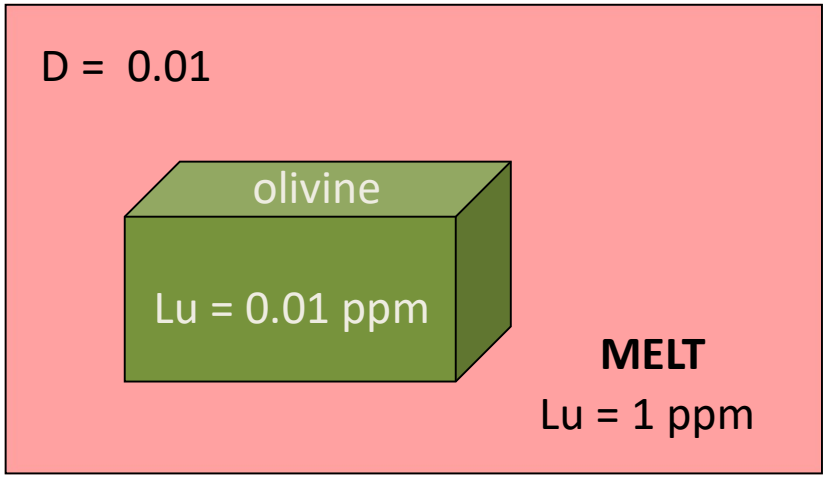
Partitioning occurs because a mineral lattice is favourable or unfavourable for a given ion, depending on the size and charge on the ion and the size and charge of the site in the lattice.

$$\text{Partition coefficient (D)} = \frac{\text{Concentration in mineral}}{\text{Concentration in melt}}$$

Compatible element: D > 1
The element prefers to sit in the mineral.
It is *compatible in the mineral*.



Incompatible element: D < 1
The element prefers to sit in the melt.
It is *incompatible in the mineral*.



PARTITION COEFFICIENTS OF REE IN DIFFERENT MINERALS IN BASALT MELTS
- Olivine, Pyroxene, Plagioclase, Garnet, Spinel, Amphibole

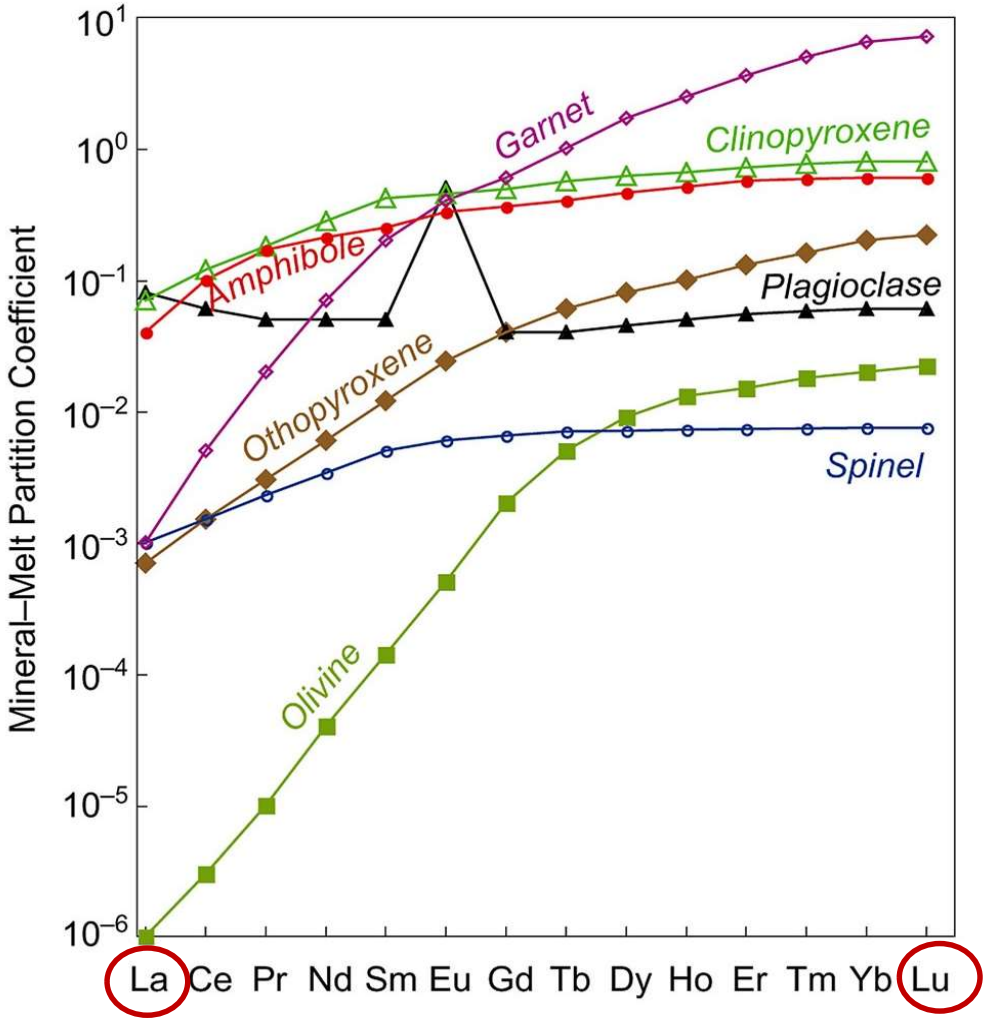
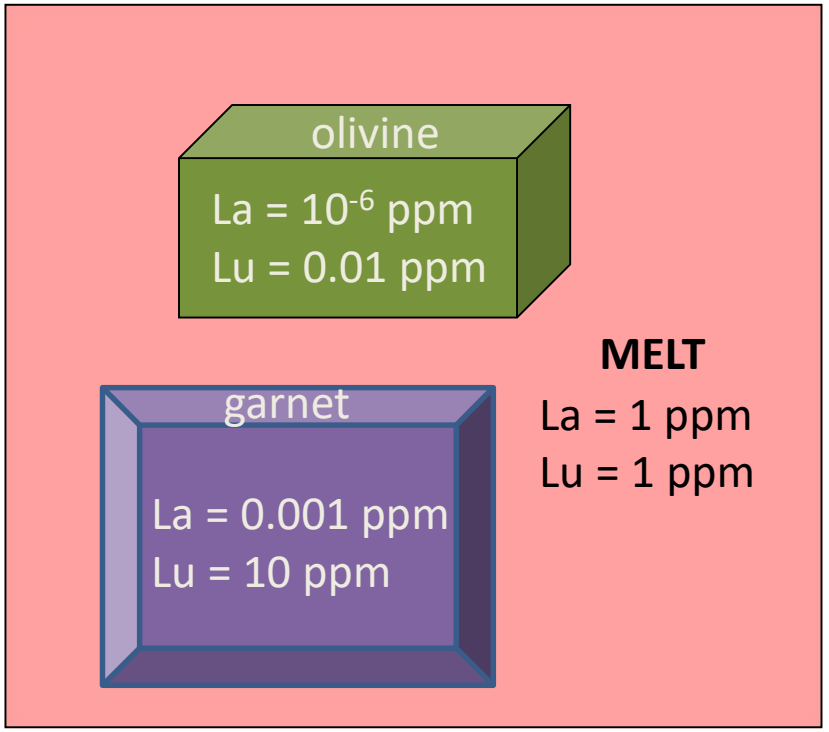


Figure 7.16 Rare earth mineral–melt partition coefficients for mafic magmas. Data from Table 7.5.

White, Geochemistry



PARTITION COEFFICIENTS OF REE IN DIFFERENT MINERALS IN BASALT MELTS
- Olivine, Pyroxene, Plagioclase, Garnet, Spinel, Amphibole

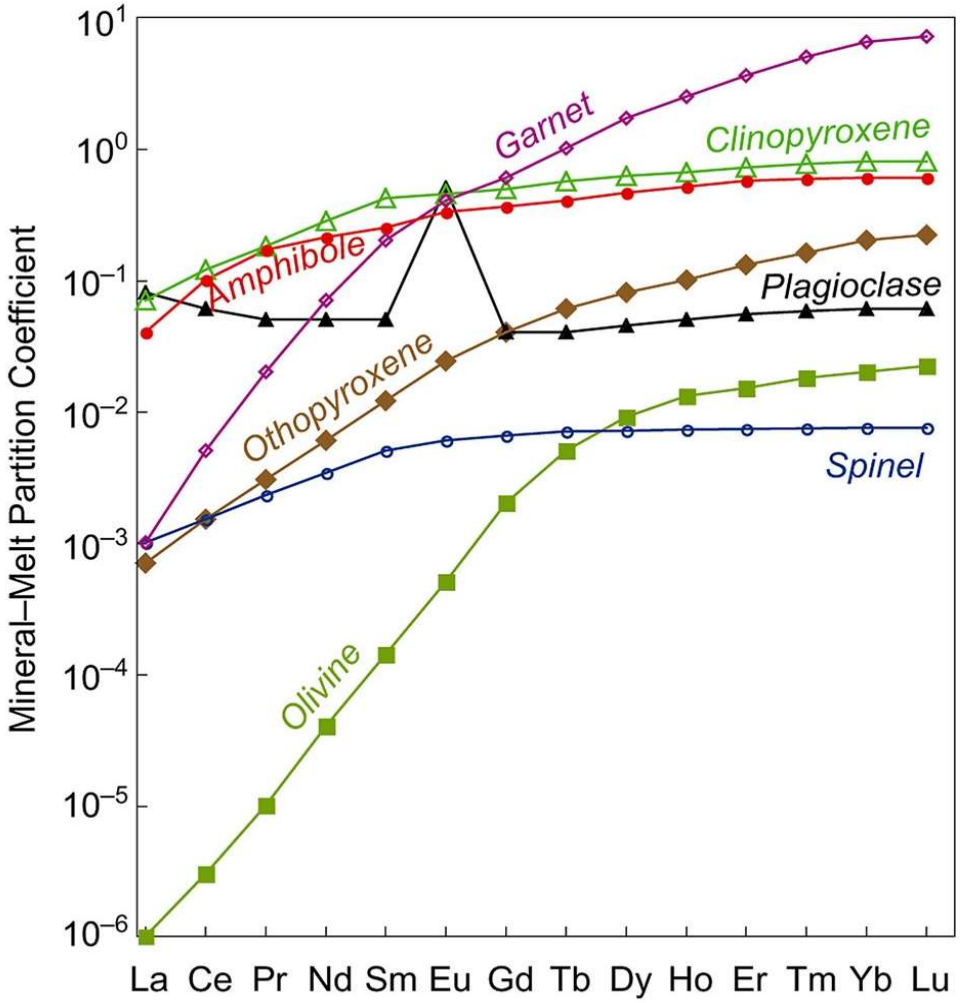


Figure 7.16 Rare earth mineral–melt partition coefficients for mafic magmas. Data from Table 7.5.

White, Geochemistry

- Patterns vary smoothly across the series because of smooth changes in ionic radii.
- Eu is an exception because it is commonly 2+ not 3+ (very clear in plagioclase)
- In most minerals, all REE are incompatible (D < 1). This means REE prefer the liquid.
- In most minerals, La is more incompatible than Lu. (La^{3+} is bigger than Lu^{3+})
- Because partition coefficients are a ratio, there is no need to normalize in order to get smooth patterns.

Chemistry of the Earth: Practical 7
Melting and the Rare Earth Elements (REE)



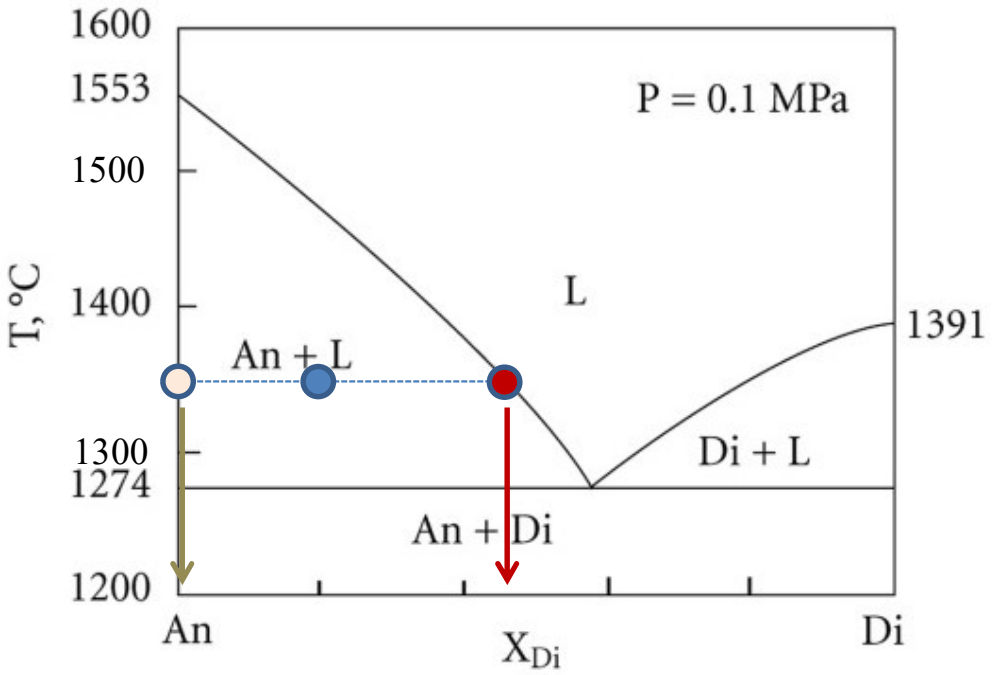
Partial Melting in the Mantle

Practical 7:
Melting and the Rare Earth Elements (REEs)

- Learn how to interpret partial melting of a rock as shown in a phase diagram
- Understand why the Rare Earth Elements (REE) are powerful trace elements that can be used to interpret geochemical processes:
    - Investigate the behaviour of REE in different minerals
    - Use the Rare Earth Elements (REE) to interpret melting processes in the mantle

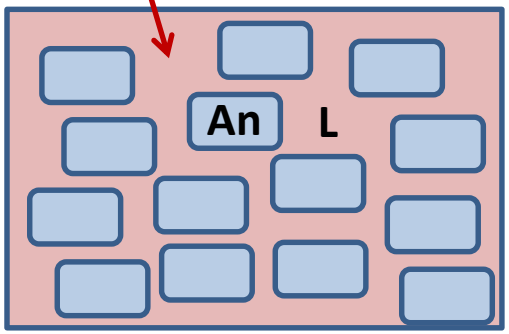
How to read a binary (two-component) phase diagram

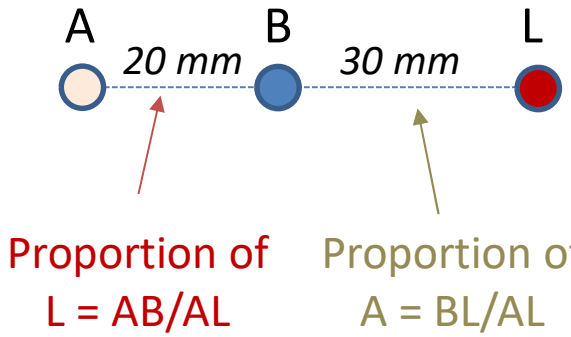
Example: Bulk composition $An_{20}Di_{80}$



At 1350 °C:
Anorthite plus Liquid

Composition of the liquid:
$X_{Di} = 0.45$
$An_{55}Di_{45}$





Lever Rule:
The proportion of anorthite is given by: BL / AL
The proportion of liquid is given by: AB / AL

Example:
Proportion of liquid = 20/50 = 40%
40% of the system is liquid, 60% is solid anorthite

PARTITION COEFFICIENTS OF REE IN DIFFERENT MINERALS IN BASALT MELTS
- Olivine, Pyroxene, Plagioclase, Garnet, Spinel, Amphibole

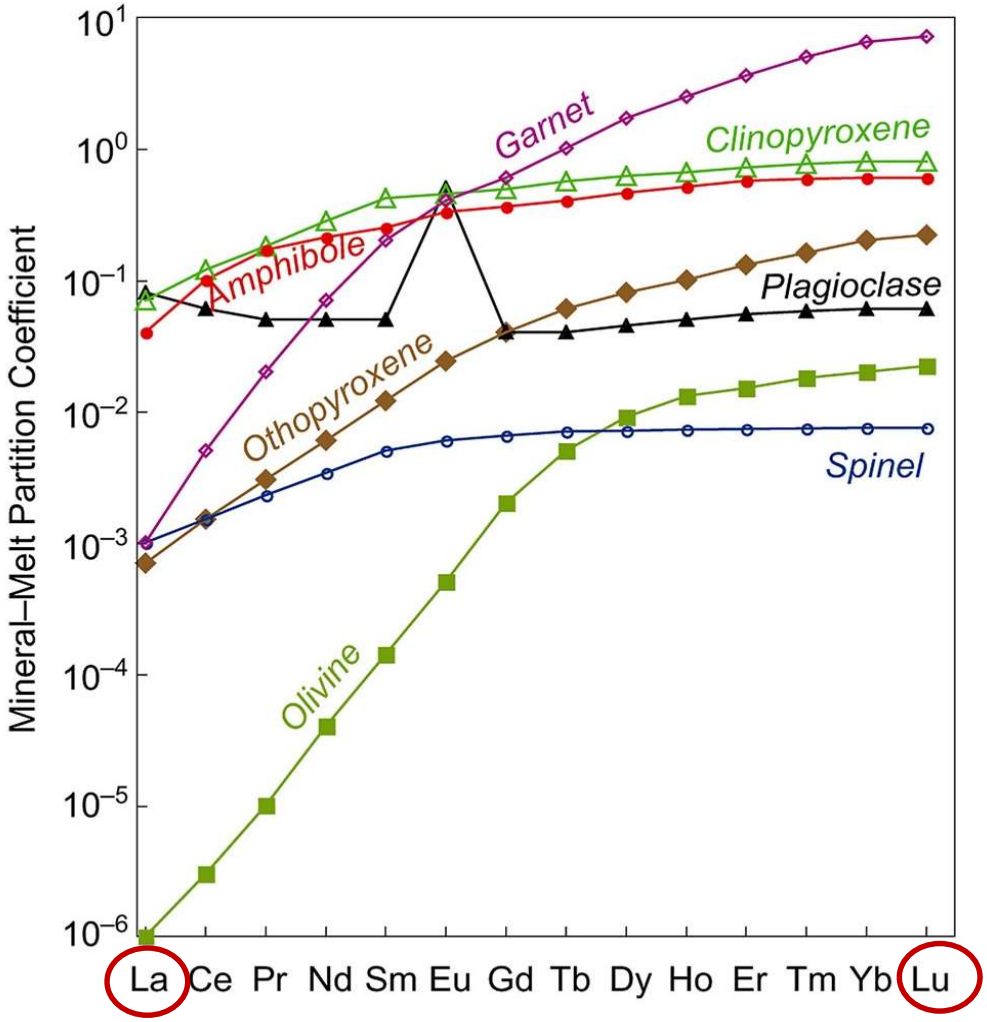
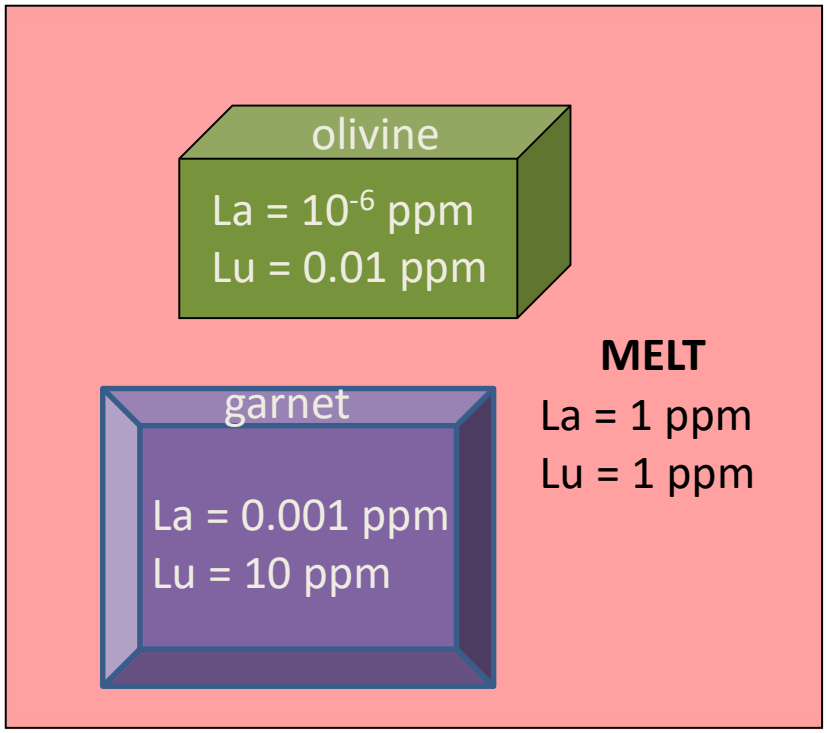


Figure 7.16 Rare earth mineral–melt partition coefficients for mafic magmas. Data from Table 7.5.

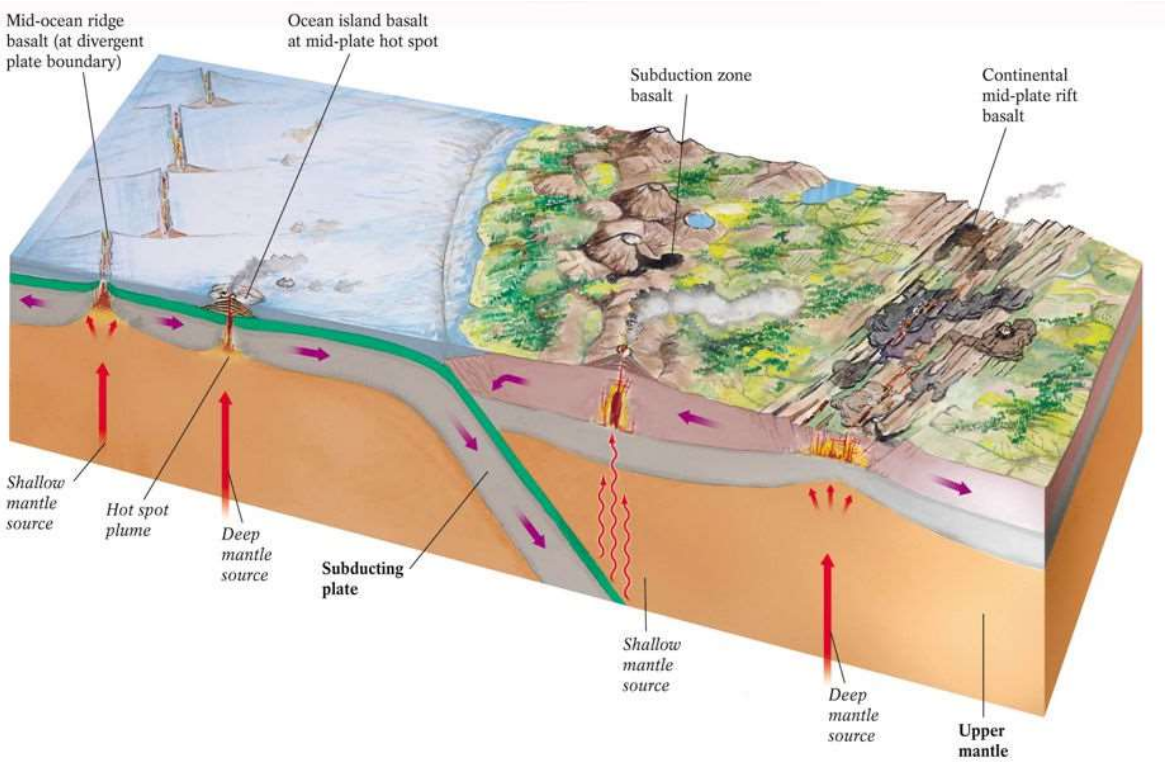
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Partition coefficient (D)

$$D_{min/melt} = \frac{\text{Concentration in mineral}}{\text{Concentration in melt}}$$



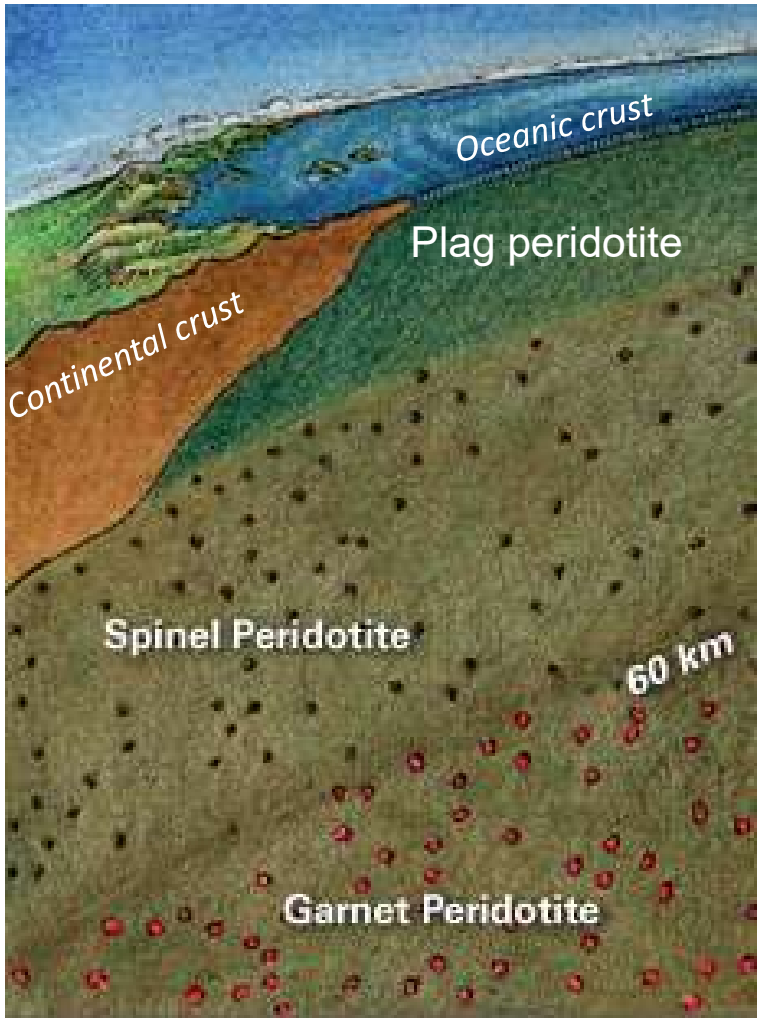
HOW ARE PARTITION COEFFICIENTS OF REE USEFUL?



When we consider the **origin of basalts**, we can use REE to:

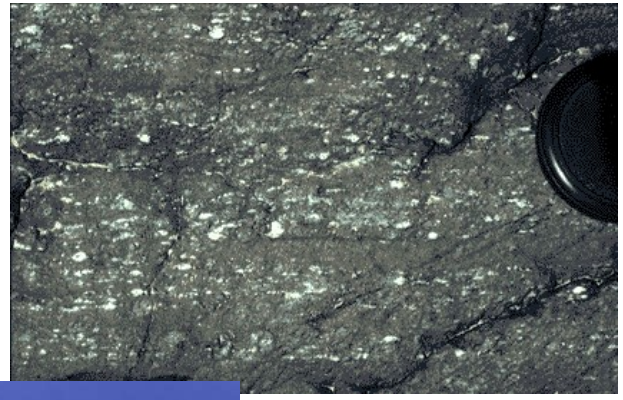
- Determine the extent of melting – did we move off the eutectic?
- Determine whether there has been any crustal assimilation as the basalt rose from the mantle – can we explain the REE abundance by mantle melting alone?
- Determine the depth of melting – what minerals were present in the mantle source?

DETERMINING THE DEPTH OF MELTING



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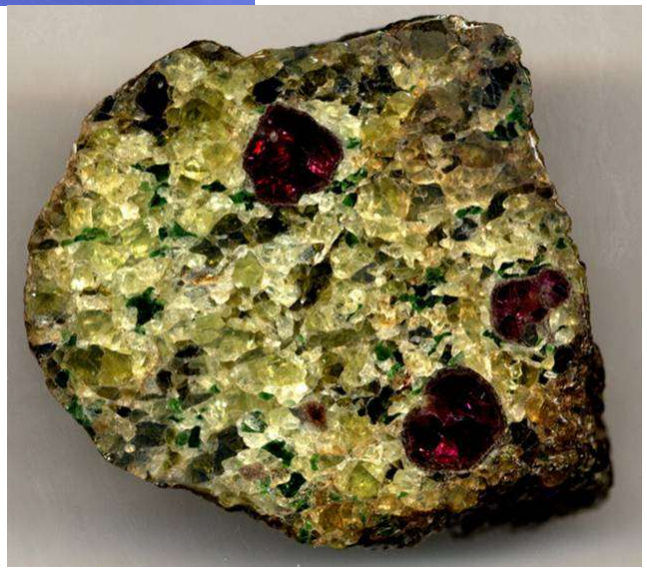
plagioclase peridotite
< 1 GPa





spinel peridotite
1-2 GPa

garnet peridotite
> 2 GPa



The REE pattern of a basalt depends on which minerals were present in the mantle source.